Fatores de qualidade de vida no trabalho de quem produz software

Achados da etapa 1: catálogo de fatores, descrição de cada um e o que cada artigo diz sobre ele

64	65	234	122 / 112
fatores	artigos com fatores	ocorrências	positivas / negativas

Base: **server-20260924-162133**. O backup congelado da etapa 2 (server-20260924-162133) é o estado depois da auditoria das citações (docs/auditoria-fatores-2026-09-24.md): só ficaram artigos primários elegíveis e com PDF, e os 14 inelegíveis apontados pela auditoria foram retirados.

Como ler. Os fatores estão agrupados nas seis categorias temáticas e, dentro de cada categoria, ordenados pelo número de artigos. Cada fator traz a descrição do catálogo e, em seguida, um bloco por artigo: a referência, a polaridade (**+ positiva**: o fator favorece o bem-estar; **– negativa**: o fator o prejudica), o termo que o próprio artigo usa e a evidência, com a localização no texto e a citação verbatim.

O nome do autor em cada bloco leva às referências completas, no fim do documento, e lá cada referência lista os fatores que sustenta. Os nomes dos fatores no sumário levam à seção do fator.

Sumário dos fatores

Art. = nº de artigos · + / − = ocorrências positivas e negativas

Tarefa e carga de trabalho	**Art.**	**+**	**−**
Flexibilidade de horário	12	12	–
Pressão de prazos	11	–	11
Carga de trabalho	10	–	10
Interrupções no trabalho	9	–	9
Conflito trabalho-família	8	–	8
Pressão por atualização	6	1	5
Ambiguidade de papel	4	–	4
Desenho do trabalho	4	–	4
Senso de realização	4	3	1
Carga cognitiva	2	–	2
Conflito de papéis	1	–	1
Técnico e ferramentas	**Art.**	**+**	**−**
Documentação interna	5	4	1
Ferramentas de IA para codificação	5	4	1
Manutenção do ambiente de desenvolvimento	5	–	5
Suporte e treinamento de TI	4	4	–
Imposição de ferramentas	3	–	3
Complexidade do código	2	–	2
Dívida técnica	2	–	2
Ergonomia visual	1	1	–
Qualidade dos requisitos	1	1	–
Processo de desenvolvimento	**Art.**	**+**	**−**
Interação com clientes	8	–	8
Adoção de métodos ágeis	1	–	1
Alinhamento das práticas ágeis	1	1	–
Estruturação do processo de trabalho	1	1	–
Gamificação no trabalho	1	–	1
Equipe e relações	**Art.**	**+**	**−**
Relações no trabalho	25	23	2
Segurança psicológica	9	8	1
Empatia no trabalho	4	4	–
Integração de novatos	3	3	–
Mentoria	3	3	–
Competição no trabalho	2	–	2
Diversidade de competências	2	2	–
Comunicação de más notícias	1	–	1
Fadiga por empatia	1	–	1
Instabilidade da equipe	1	–	1

Organizacional	Art.	+	–
Apoio do gestor	12	11	1
Insegurança no emprego	6	1	5
Inclusão e representatividade	5	5	–
Cultura autoritária	4	–	4
Monitoramento no trabalho	4	2	2
Reconhecimento no trabalho	4	4	–
Iniciativas superficiais de bem-estar	3	–	3
Senso de propósito	3	3	–
Justiça organizacional	2	–	2
Acomodações para neurodivergência	1	1	–
Clima de aprendizagem	1	1	–
Conformidade e sigilo de dados	1	–	1
Cultura organizacional generativa	1	1	–
Estigma da neurodivergência	1	–	1
Estigma do uso de substâncias	1	–	1
Oportunidades de carreira	1	1	–
Política antidrogas	1	–	1
Preparação organizacional para crises	1	1	–
Silêncio organizacional	1	–	1
Treinamento de soft skills	1	1	–

Individual	Art.	+	–
Alinhamento do uso do tempo	5	4	1
Práticas de mindfulness	4	4	–
Qualidade do sono	3	3	–
Contatos sociais fora do trabalho	1	1	–
Coping focado na emoção	1	–	1
Coping focado no problema	1	1	–
Divulgação da neurodivergência	1	1	–
Fenômeno do impostor	1	–	1
Uso de substâncias psicoativas	1	1	–

Tarefa e carga de trabalho

11 fatores · 36 artigos distintos

Flexibilidade de horário

12 artigos · 12 positivas · 0 negativas

> Poder escolher quando e de onde trabalhar: home office, híbrido, começar mais tarde, ajustar o dia sem pedir permissão, não gastar horas em deslocamento. É um dos itens mais citados espontaneamente como o que mais melhora a vida de trabalho, e sua retirada é sentida como perda concreta.

\+ **Ågren et al. (2022) — Agile software development one year into the COVID-19 pandemic**
Empirical Software Engineering · termo no artigo: "perceived having more flexibility and freedom since they can work from any place they want"
Seção 4.4.1 (7 entrevistados; 45% do questionário relataram aumento do equilíbrio trabalho-vida): "Several interviewees perceived having more flexibility and freedom since they can work from any place they want and manage their time more easily when working remotely".

\+ **Juarez-Ramirez et al. (2022) — Assessing Wellbeing and Soft Skills of Software Developers and Students in the Context of Working from Home during the COVID-19 era**
Proceedings - 2022 10th International Conference in Software Engineering Research and Innovation, CONISOFT 2022 · termo no artigo: "maintain work schedule"
Seção V.A (dificuldades no trabalho remoto): "In the case to maintain work schedule, 42% of developers had difficulties, while 76% of students have it. This is a warning to consider in both cases, but especially for students".

\+ **Neumann et al. (2022) — The Covid 19 Pandemic and its Effects on Agile Software Development**
ACM International Conference Proceeding Series · termo no artigo: "certain team members chose to work partially during a different time in the day"
Seção 4.2: "An increase in remote work would not mean 100% of the time, as 17 out of 19 interviewees favor a combined form of work". Seção 4.3: "Our results show that certain team members chose to work partially during a different time in the day when they felt the most undisturbed. [...] Employees could attend appointments which would not be possible when working on-site".

\+ **Ragkhitwetsagul et al. (2025) — The Impact of COVID-19 and Remote Work on Software Development in Thailand**
Proceedings - 9th International Conference on Information Technology, InCIT 2025 · termo no artigo: "Flexible work hours"
RQ3 (survey, 194 devs tailandeses; Tabela IV: B1 93%/92%, B3 83%/77%): "the Thai developers received the benefits of less commute time, spending less money, having flexible work hours, and being closer to their families".

\+ **Romano et al. (2024) — MOOD: Mindfulness fOr sOftware Developers**
International Symposium on Empirical Software Engineering and Measurement · termo no artigo: "the autonomy in the organization of working time"
Seção 5 (3 grupos focais): "As for the job resources, we found (among other things) that software developers appreciate the autonomy in the organization of working time (e.g., deciding whether to work in the office or remotely)".

\+ **Russo et al. (2024) — Understanding Developers Well-Being and Productivity: A 2-year Longitudinal Analysis during the COVID-19 Pandemic**
ACM Transactions on Software Engineering and Methodology · termo no artigo: "hybrid work setting"
Seção 4.5 (análise temática, Wave 6, 84.5% de sentimento positivo): "several respondents stated their belief that a hybrid work setting provides a tremendous productivity and well-being booster". Seção 4.3 (interação significativa, $p = .043$): "When the company preferred employees to work from the office, there was a positive association between their own preference for working remotely and intention to leave jobs, $B = .02$, $SE = .01$, $p = .05$".

+ **Santos et al. (2024) — Hybrid Work Well-Being: Software Professionals Finding Equilibrium**
IEEE Software · termo no artigo: “hybrid work”
Seção 4.1 (545 profissionais de software, estudo de caso): “hybrid work has a positive impact on four out of the five dimensions, specifically Positive Emotions, Engagement, Relationships, and Accomplishment”. Tabela I (P006): “My experience has been great because I can work on a more flexible schedule. Since I have morning classes, I can work in the afternoon and even a bit at night, which benefits me greatly as I can’t miss those classes.”

+ **Siitonen et al. (2025) — Coping with technostress in the software industry: Coping strategies and factors underlying their selection**
Journal of Systems and Software · termo no artigo: “having a flexible work schedule”
Seção 4.1.1: “Organizational factors that helped with disengagement efforts were an accepting atmosphere regarding stress, being allowed or even encouraged to take breaks in case of exhaustion, and having a flexible work schedule”.

+ **Smite et al. (2025) — Dual Effects of Hybrid Working on Performance: More Work Hours or More Work Time**
Lecture Notes in Business Information Processing · termo no artigo: “the lack of commute time that led to having more work time”
Seção 3 (192 respondentes): “Among these, we found, for example, positive effects, such as the lack of commute time that led to having more work time”. Seção 4: “We found increased work hours to be associated with a more efficient use of time (contributing to having more work time)”; “the lowest increase in work hours was reported by those only in the office 1–2 days a week. This may be due to having more control over work schedule, and fewer interruptions when performing individual tasks”.

+ **Upeksha et al. (2025) — The Influence of Generative AI on work-life balance among female software professionals in Sri Lanka**
2025 5th International Conference on Advanced Research in Computing: Converging Horizons: Uniting Disciplines in Computing Research through AI Innovation, ICARC 2025 - Proceedings · termo no artigo: “hybrid or flexible work arrangements”
Seção 4.2.2 (tema Flexibility, 60% das menções; R09): “Most companies now offer hybrid or flexible work arrangements, allowing you to manage your time around personal commitments”.

+ **Weerasekara et al. (2020) — ICT supported occupational stress management: The views of hr managers in Sri Lankan software companies**
Proceedings of the 12th IADIS International Conference e-Health 2020, EH 2020 - Part of the 14th Multi Conference on Computer Science and Information Systems, MCCSIS 2020 · termo no artigo: “flexible working hours and working from home facility”
Seção 5.1.3: “They also provided flexible working hours and working from home facility to let the employees feel more relaxed at work”. Seção 5.2.1 (HR10): “When there’s delivery/deployment, the stress level goes high [...] That is why we create a space for them to relax, provide flexible working hours and flexibility in everything”.

+ **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “desire for flexible work hours”
Seções 4.2.1 e 4.3.1: horários flexíveis sem controle constante e comunicação assíncrona sustentaram a autonomia (P17, P20, P23, P24) — “participants highlighted the importance of flexible hours without micromanagement”.

Pressão de prazos

11 artigos · 0 positivas · 11 negativas

> Prazo apertado, urgência permanente e prioridade que muda no meio do caminho. Diferente de carga de trabalho: aqui o problema é o relógio e a instabilidade do alvo, não a quantidade. Inclui estimativa tratada como compromisso e entrega iterativa que virou cobrança contínua.

− **Benlian et al. (2025) — Team-Enacted Use vs. Developer-Needed Use of Agile Practices: How Perceptual (In-)Congruence and Team Feedback-Seeking Shape Developer Well-Being**

Information Systems Research · termo no artigo: "Project priorities and deadlines often shift rapidly"

Entrevistas de follow-up (I26): "Project priorities and deadlines often shift rapidly [in our project], and so causing the team to focus more on delivery than on thorough code reviews or retrospectives".

− **Chattopadhyay et al. (2021) — Developers Who Vlog: Dismantling Stereotypes through Community and Identity**

Proceedings of the ACM on Human-Computer Interaction · termo no artigo: "aggressive focus on shipping elements"

Seção 5.2.5 (vlogs de desenvolvedores de multinacionais): "One developer from a multinational company working on a fully developed product discussed the "aggressive focus on shipping elements""; [V34]: "high pay means [a requirement for] a high accomplishment. It also means a high pressure" which creates and accumulates over the years".

− **Graziotin et al. (2017) — On the unhappiness of software developers**

ACM International Conference Proceeding Series · termo no artigo: "Time pressure"

Tabela 2 e Seção 4 (1.318 respondentes): Time pressure é a causa nº 2 (152 menções) — "the respondents consider that high time pressure they feel, often generated by "unrealistic", "unjustified" and "crazy" deadlines, will almost surely push them into very unhappy states".

− **Madampe et al. (2023) — The Emotional Roller Coaster of Responding to Requirements Changes in Software Engineering**

IEEE Transactions on Software Engineering · termo no artigo: "whether or not I had enough time to make the change"

Seção 4.4 (P150, Developer/Tester): "I felt anxious because I was worried about whether or not I had enough time to make the change and complete the project on time". Em contraste, Seção 4.4.1: "If an RC fits the project timeline, practitioners feel high2 and high1 emotions, thus making it the way to direct emotion regulation towards high pleasure".

− **Madampe et al. (2024) — Addressing Bad Feelings in Agile Software Project Contexts: Considering Team Welfare and Developer Mental Health**

IEEE Software · termo no artigo: "Deadline approached"

Seção "Developer productivity and their emotions" (Agile Coach/Scrum Master e Developer): "Deadline approached: None of the team members could solve the unresolved issues"; "The above emotions happened to me when the deadline of the project approached. At the last day of the release there were some issues which couldn't be sorted out by anyone".

− **Obi et al. (2025) — Identifying Factors Contributing to "Bad Days" for Software Developers: A Mixed-Methods Study**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "rapidly changing team priorities"

Seção IV.A.1b (22 entrevistas; Fig. 4): "Issues around unclear project ownership, lack of documentation and knowledge sharing, and rapidly changing team priorities were other common factors that caused 'bad days' for developers".

- **Olsson et al. (2021) — Measuring affective states from technical debt: A psychoempirical software engineering experiment**
  Empirical Software Engineering · termo no artigo: “time pressure to do something”
  Tema Apprehension (40 profissionais de 12 empresas): “the psychological toll can also appear in the presence of tight schedules. As one practitioner put it, “If you have time pressure to do something, and then you also know that you’re in—I mean, “this is gonna be hard to test. And to deliver it in time is gonna be tough.” Then it’s super stressful. But if you don’t have that pressure again, then it’s easier again””. A pressão de prazo amplifica o dano emocional da dívida técnica.

- **Romano et al. (2024) — MOOD: Mindfulness fOr sOftware Developers**
  International Symposium on Empirical Software Engineering and Measurement · termo no artigo: “Time pressure”
  Seção 5 (3 grupos focais com desenvolvedores): “Time pressure is considered an important stress factor: deadlines often overlap one another so that developers have to accomplish more work at the same time, adding further difficulty in time management; moreover, most work is categorized as ‘urgent’”.

- **Weerasekara et al. (2020) — ICT supported occupational stress management: The views of hr managers in Sri Lankan software companies**
  Proceedings of the 12th IADIS International Conference e-Health 2020, EH 2020 - Part of the 14th Multi Conference on Computer Science and Information Systems, MCCSIS 2020 · termo no artigo: “the stress level goes high”
  Seção 5.2.1 (HR10): “When there’s delivery/deployment, the stress level goes high because they have to put their full strength, work in full time, work overnight, deal with customers and managers. Those days their stress level is high, and that we know”.

- **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**
  Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “urgent deadlines”
  Seção 4.1.1 (14 entrevistas com engenheiros de software): “participants described situations beyond their control, including urgent deadlines and changes made by stakeholders that resulted in wasted time on development work”. Seção 4.1.2 (P1): “had no choice but to try and like push through it”.

- **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**
  Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “the pressure from task urgency and deadlines”
  Seção 4.2.1 (P2): “the pressure from task urgency and deadlines could lead to feeling a lack of competence”.

Carga de trabalho

10 artigos · 0 positivas · 10 negativas

> Volume e pressão: quantidade de tarefa na fila, hora extra recorrente, escopo que cresce durante a execução, várias frentes ao mesmo tempo. É quantidade, não dificuldade nem prazo. Aparece como noite e fim de semana ocupados e como sensação de nunca zerar, e é dos fatores mais associados a esgotamento.

- **Arriel et al. (2026) — Understanding developer well-being: measuring mental health and productivity in software teams**
  Empirical Software Engineering · termo no artigo: “work overload and excessive pressure”
  Seção 6.1.3: sobrecarga e pressão por resultado empurram para burnout e derrubam o desempenho no longo prazo (P14) — “work overload and excessive pressure were repeatedly cited as threats to both mental health and productivity”.

- **Chattopadhyay et al. (2021) — Developers Who Vlog: Dismantling Stereotypes through Community and Identity**
  Proceedings of the ACM on Human-Computer Interaction · termo no artigo: "long stressful working hours"
  Seção 5.2.5 (análise de 130 vlogs de desenvolvedores): "Likewise, long stressful working hours can lead to exhaustion at the end of the day, as a developer states "now I had a really kind of busy morning, I'm kind of mentally drained" [V24]". Finding 13: "Software developers lead a stressful work-life, with long hours, strict deadlines, and many moving software components requiring constant coordination with multiple teams involved in the project". [V34]: "After years of the stress for days and a night sitting long hours in front of the desks, it creates tension, headaches, low back pain, dry eyes, and obesity".

- **Chilton et al. (2010) — Performance and Strain Levels of IT Workers Engaged in Rapidly Changing Environments: A Person-Job Fit Perspective**
  Data Base for Advances in Information Systems · termo no artigo: "needs for quantitative workload"
  Seção Results (124 desenvolvedores OO de 8 empresas norte-americanas em transição para orientação a objetos): "we can sustain hypotheses 2a through 2c and conclude that the congruence of a worker's beliefs about his or her need for quantitative workload, supervisory support and job ambiguity in an ideal job with what is supplied on a new job relative to an old job is negatively related to strain". Ainda em Results: "we see that the greatest amount of variance is explained by quantitative workload. This is not an unexpected result since it no doubt includes time that the incumbent worker may have to devote to learning and applying the new technology".

- **Espina (2026) — Software Engineering Occupational Stress Factors and Work Performance**
  ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management · termo no artigo: "handling multiple roles"
  Seção 3.4 (M=3.25): "workload remains one of the most persistent sources of occupational stress among software developers".

- **Leme et al. (2024) — Mental Health and Productivity in Software Development: A Study with the Bravo Central Platform**
  ACM International Conference Proceeding Series · termo no artigo: "estratégias para melhorar a distribuição de tarefas"
  Seção 5.1 (6 membros do Studio Bravarda, 16 semanas de acompanhamento com a plataforma Bravo Central): "A equipe reconheceu que um excesso de tarefas para alguns membros pode levar a sobrecarga e, eventualmente, impactar negativamente na saúde mental, diminuindo a produtividade geral da equipe. A reflexão coletiva resultou em estratégias para melhorar a distribuição de tarefas". Seção 5.2: "Cargas de trabalho flutuantes: períodos intensos de trabalho, prazos apertados ou fases críticas do projeto".

- **Madampe et al. (2023) — A Framework for Emotion-Oriented Requirements Change Handling in Agile Software Engineering**
  IEEE Transactions on Software Engineering · termo no artigo: "the effort required to work on the RC is high"
  Seção V.B: "Low2 emotions tend to be felt when an RC is large"; "when the effort required to work on the RC is high, low2 emotions tend to be felt by practitioners". βP10: "Large new features being added, that necessitate large rewrites as they were well beyond the initial scope".

- **Micallef e Porter (2019) — Towards detecting and managing information anxiety in the ICT industry**
  Proceedings of the International Conference on Software Engineering and Knowledge Engineering, SEKE · termo no artigo: "information overload"
  Seções V.B e V.C (18 profissionais de TIC malteses acompanhados por um mês com autorrelatos diários de ansiedade informacional): "All eight of these participants indicated that information overload was the main cause of anxiety"; "there is a significant perception amongst software engineers in our cohort that they have too much information to process and feel overwhelmed by it". Seção V.D: "The data indicates that as participants gain experienced, they suffer less from information overload".

- **Russo et al. (2023) — Satisfaction and performance of software developers during enforced work from home in the COVID-19 pandemic**
  Empirical Software Engineering · termo no artigo: "I had a lot of pressures I could do without"
  Seção 4.3 (modelos lineares mistos, N=290): "Across all activities, activity satisfaction was negatively predicted by conflicts and pressure". Item da medida: "I had a lot of pressures I could do without when working".

Smite et al. (2025) — Dual Effects of Hybrid Working on Performance: More Work Hours or More Work Time
Lecture Notes in Business Information Processing · termo no artigo: "an increase in work hours"

Seção 3 e 4 (192 respondentes): "the other half of respondents (49%) report an increase in work hours"; "we conclude that these concern job pressure and individual habits related to increased connectivity". Abstract: "we recommend managers to discuss and address the concerning habits of employees caused by increased connectivity and inability to stop working, before these lead to burnout and disturbances in the work/life balance".

Upeksha et al. (2025) — The Influence of Generative AI on work-life balance among female software professionals in Sri Lanka
2025 5th International Conference on Advanced Research in Computing: Converging Horizons: Uniting Disciplines in Computing Research through AI Innovation, ICARC 2025 - Proceedings · termo no artigo: "working for long hours especially in the evenings"

Seção 4.2.10 (R03): "I have to work beyond schedules for hours and it's fairly common in this system engineering role"; (R07): "the company expects us to use AI tools more frequently and to work extra hours to develop solutions faster".

Interrupções no trabalho

9 artigos · 0 positivas · 9 negativas

> Tudo que quebra bloco de foco: agenda picada por reuniões curtas ao longo do dia, mensagem com expectativa de resposta imediata, ambiente barulhento, chamado fora de hora. Programar exige carregar contexto na cabeça, e a interrupção derruba esse contexto — o custo é muito maior que o tempo da interrupção.

Brandebusemeyer et al. (2025) — Wearables to measure developer experience at work
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "too many interruptions"

Seção IV.F (RQ4): excesso de reuniões e de interrupções estão entre os dez maiores desafios, e as melhorias pedidas incluem espaço silencioso e tempo de foco acordado — "too many meetings, finding information of documentation, too many interruptions".

Kuutila et al. (2021) — Individual differences limit predicting well-being and productivity using software repositories: a longitudinal industrial study
Empirical Software Engineering · termo no artigo: "Private messages from the chat system were seen as interrupting"

Seção 4.4 (Role of Instant Messaging): "One negative consequence of the chat system was mentioned by a single respondent. Private messages from the chat system were seen as interrupting, as they felt there was a higher urgency to respond since a response was demanded specifically from them". Em contraste, o chat público trouxe "better focus without interruptions in the shared working space".

Meyer et al. (2021) — Enabling good work habits in software developers through reflective goal-setting
IEEE Transactions on Software Engineering · termo no artigo: "Improve Focus: Reduce Distractions and Interruptions"

Seção 4.2 (52 desenvolvedores profissionais em auto-reflexão diária; 33 estabeleceram esta meta): "developers aim to limit their exposure to aspects that reduce their focus, to be able to concentrate for longer chunks of time on planned work. Specifically, developers' goals primarily aim to reduce the amount of external interruptions from co-workers and self-distractions". P41: "Reduce meeting time or club meetings back to back to have bigger focus time blocks on my calendar".

Obi et al. (2025) — Identifying Factors Contributing to "Bad Days" for Software Developers: A Mixed-Methods Study
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "Interruptions and Randomizations"

Seção IV.A (Tabela II, média 4.02; Fig. 4): "You have more than half your day filled with meetings".

- **Ragkhitwetsagul et al. (2025) — The Impact of COVID-19 and Remote Work on Software Development in Thailand**
  Proceedings - 9th International Conference on Information Technology, InCIT 2025 · termo no artigo: "More distractions or interruptions"
  Seção IV.E.2 (Tabela V: C11 importância 69,52%; C10 prevalência 65,79%): "and C10 Too many meetings (65.79%)".

- **Russo et al. (2021) — Predictors of well-being and productivity among software professionals during the COVID-19 pandemic – a longitudinal study**
  Empirical Software Engineering · termo no artigo: "distinguishing between distraction and office set-up"
  Seção 5.4 (análise de replicação com os dados de Ralph et al.): "ergonomics distractions was more strongly correlated with well-being, r = .25, and productivity, r = .29, than was ergonomics-other, rs = .19 and .19, respectively. This suggests that our findings replicate those of Ralph et al. and emphasize the importance of distinguishing between distraction and office set-up". Sobre o próprio survey: office-setup "was positive but not significantly associated with well-being, r = .14, and productivity, r = .10".

- **Russo et al. (2024) — Understanding Developers Well-Being and Productivity: A 2-year Longitudinal Analysis during the COVID-19 Pandemic**
  ACM Transactions on Software Engineering and Methodology · termo no artigo: "distractions at home"
  Seção 4.3 (regressão com variáveis da Wave 6): "Of the 14 predictors, only solidarity, B = .18, SE = .05, p < .001, distractions at home, B = −.17, SE = .08, p = .04 [...] were associated with job satisfaction".

- **Santos et al. (2024) — Hybrid Work Well-Being: Software Professionals Finding Equilibrium**
  IEEE Software · termo no artigo: "interruptions and distractions often occur due to a crowded workspace"
  Seção 4.1: "software professionals reported heightened concentration in this environment. Notably, they attributed this to the new challenges faced in the office environment, where interruptions and distractions often occur due to a crowded workspace". Tabela I (P100): "The benefits are the silence in the environment. In practice, the office is full of distractions."

- **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**
  Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "pressure to like always be on for Slack"
  Seção 4.1.1 (P1): "Some participants talked about "pressure to like always be on for Slack" (P1), self-doubts, and concerns over performance and job security as contributing factors to their poor mental wellbeing".

Conflito trabalho-família

8 artigos · 0 positivas · 8 negativas

> Trabalho invadindo a vida pessoal e vice-versa: notebook aberto depois do jantar, reunião em fuso alheio, ausência de um fim claro do dia, casa sem lugar adequado para trabalhar. Piorou muito com home office, e pesa mais para quem mora com crianças pequenas.

- **Espina (2026) — Software Engineering Occupational Stress Factors and Work Performance**
  ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management · termo no artigo: "Work-Family Interface"
  Seção 3.4: segundo maior estressor (M=3.31), puxado por pensar no trabalho mesmo em casa (M=3.8) e por trabalhar de madrugada ou em fins de semana (M=3.26) — "difficulty maintaining work–life boundaries are a key stressor for software professionals".

- **Meem e Johnson (2025) — Investigating the Impact of AI-Assisted Tools on Software Practitioner Well-Being**
  CHIWORK 2025 - Adjunct Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work · termo no artigo: "blurred work-life boundaries"
  Seção 4.1 (41 praticantes): "a majority of respondents (52%) admitted to occasionally working beyond regular hours due to AI tool usage, with a smaller yet noteworthy group (17%) frequently extending their workdays".

- **Meyer et al. (2021) — Enabling good work habits in software developers through reflective goal-setting**
  IEEE Transactions on Software Engineering · termo no artigo: “Better Detach from Work”

  Seção 4.4 (meta G4 Maintain Work-Life Balance; 7 desenvolvedores): “Outside of work, developers want to be able to enjoy their spare time more and regenerate for work, by better detaching from work and not thinking about it all the time”. P11: “Don’t think about work outside of work. Try to be satisfied with the daily work and come back fully rested”. Estratégias em 4.4: “reducing their work hours (S24) [...] and by working more regular hours (S25) and avoiding night shifts or weekend work”.

- **Ragkhitwetsagul et al. (2025) — The Impact of COVID-19 and Remote Work on Software Development in Thailand**
  Proceedings - 9th International Conference on Information Technology, InCIT 2025 · termo no artigo: “Lack of work-life boundary”

  RQ3 (Tabela V: C2, prevalência 70%): “they faced challenges of missing social interactions, a lack of work-life boundary, less awareness of colleagues work, difficulty in communication with colleagues, and having too many meetings”.

- **Ralph et al. (2020) — Pandemic programming: How COVID-19 affects software developers and how their organizations can help**
  Empirical Software Engineering · termo no artigo: “People who live with small children have significantly less ergonomic home offices”

  Seção 5.6 (achados exploratórios, efeitos diretos no SEM): “People who live with small children have significantly less ergonomic home offices. This is not surprising because the ergonomics scale included items related to noise and distractions”. Abstract: “women, parents and people with disabilities may be disproportionately affected”.

- **Romano et al. (2024) — MOOD: Mindfulness fOr sOftware Developers**
  International Symposium on Empirical Software Engineering and Measurement · termo no artigo: “work-life conflicts”

  Seção 5 (3 grupos focais): “software developers are subject to work-life conflicts such as difficulties in detaching from the problems they tackle at work or meetings arranged outside working hours”.

- **Smite et al. (2025) — Dual Effects of Hybrid Working on Performance: More Work Hours or More Work Time**
  Lecture Notes in Business Information Processing · termo no artigo: “increased connectivity”

  Seção 3 (192 respondentes do departamento de engenharia de software da Storebrand, fintech norueguesa, 48% de resposta): “the most frequent reason contributing to having more or longer work hours was increased connectivity, which related to higher mobilisation and lower threshold for contacting the colleagues but also a more blurry boundaries between work and private life. Descriptions in this category were sometimes alarming, indicating that people sacrifice their free time and privacy for performing the work”. Recomendação na Seção 4: “to implement a “hard stop” practice-a preset time for logging off or turning off the computer”.

- **Upeksha et al. (2025) — The Influence of Generative AI on work-life balance among female software professionals in Sri Lanka**
  2025 5th International Conference on Advanced Research in Computing: Converging Horizons: Uniting Disciplines in Computing Research through AI Innovation, ICARC 2025 - Proceedings · termo no artigo: “blurred boundaries between work and personal life”

  Conclusão: “challenges such as increased workload expectations, blurred boundaries between work and personal life, and insufficient organizational policies on AI usage can detract from these benefits, especially female employees who may already be dealing with particular work-life demands”.

Pressão por atualização

6 artigos · 1 positiva · 5 negativas

> Sentir que precisa estudar sem parar para não ficar obsoleto: stack nova, linguagem desconhecida no meio do projeto, ferramenta que muda de ano em ano. Aparece como cobrança contínua e difusa, muitas vezes fora do horário de trabalho, e como medo real de perder empregabilidade.

\+ **Khalid et al. (2022) — Relationship between Organization's Physical and Psychological Environment and Employees' Mental Satisfaction: An Empirical Analysis of Employee Turnover in Software Industry**

2022 16th International Conference on Open Source Systems and Technologies, ICOSST 2022 - Proceedings · termo no artigo: "better learning"

Seção V.C (respostas dos empregados sobre motivo para sair): "The 33% of employees who wanted to quit for better learning is an indication of the fact that the decision making power was restricted to a certain group in most of the cases. This restriction was, in turn, hindering the people in the lower part of hierarchy to experience new things and to contribute as per their full potential".

– **Chilton et al. (2010) — Performance and Strain Levels of IT Workers Engaged in Rapidly Changing Environments: A Person-Job Fit Perspective**

Data Base for Advances in Information Systems · termo no artigo: "must constantly re-tool to keep up with the frequently changing IT environment"

Seção Building the Hypotheses: "It is clear that IT personnel often find themselves on a steep learning curve as they must constantly re-tool to keep up with the frequently changing IT environment. This in turn, has negatively affected IT productivity and performance". Seção Building the Hypotheses (mecanismo sobre carga): "developers must learn to use and assimilate the new technology. In the early stages of use, developers may make mistakes due largely to ignorance and inexperience. [...] This means that developers may feel the need to work harder and/or longer".

– **Espina (2026) — Software Engineering Occupational Stress Factors and Work Performance**

ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management · termo no artigo: "Fear of Obsolescence"

Seção 3.1 (M=3.54): "technological change and continuous learning demands are major contributors to occupational stress among software development professionals".

– **Ribeiro (2022) — Understanding the relationships between the perceptions of burnout and instability in Software Engineering**

ACM International Conference Proceeding Series · termo no artigo: "instabilidade tecnológica"

Seção 2.2 (411 respondentes): "a instabilidade tecnológica é causada pela introdução de um novo elemento tecnológico (novas linguagens, frameworks, API, modificação em software de terceiros que afeta o seu desenvolvimento de software...) ou por uma turbulência tecnológica". Resultado: relação positiva e significativa com exaustão e cinismo; "Para a dimensão da eficácia, ela foi negativa e fraca com as instabilidades tecnológicas e da equipe".

– **Siitonen et al. (2025) — Coping with technostress in the software industry: Coping strategies and factors underlying their selection**

Journal of Systems and Software · termo no artigo: "working with unfamiliar programming languages"

Seção 4.1.3 (110 respondentes) e Tabela 4: "Insecurity and doubt caused by insufficient IT use skills and/or training"; "the respondents feeling forced to use these CSs because they lacked the knowledge to resolve the incident (e.g., working with unfamiliar programming languages)".

– **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**

Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "constant pressure to learn new skills and tools"

Seção 4.1.1 (P5, P9, P21): "This ongoing pressure to learn new skills and tools challenged their relatedness".

Ambiguidade de papel

4 artigos · 0 positivas · 4 negativas

> Não saber o que se espera de você: meta escrita de forma vaga, projeto sem dono claro, prioridade que ninguém confirma, entrega cujo critério de pronto aparece só na revisão. Consome energia em adivinhação e deixa a pessoa insegura sobre estar fazendo o certo, o que é diferente de ter trabalho demais.

− **Brandebusemeyer et al. (2025) — Wearables to measure developer experience at work**
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: “poorly defined goals”
Seção IV.F (RQ4) e Tabela III: “lack of team/organization vision and poorly defined goals impact more than one-third of the participants to a great extent”.

− **Chilton et al. (2010) — Performance and Strain Levels of IT Workers Engaged in Rapidly Changing Environments: A Person-Job Fit Perspective**
Data Base for Advances in Information Systems · termo no artigo: “needs for job ambiguity”
Seção Results (124 desenvolvedores OO): as hipóteses 1c e 2c são sustentadas — “the congruence of a worker’s beliefs about his or her need for quantitative workload, supervisory support and job ambiguity in an ideal job with what is supplied on a new job relative to an old job is positively related to performance” e “negatively related to strain”.

− **Espina (2026) — Software Engineering Occupational Stress Factors and Work Performance**
ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management · termo no artigo: “Role Ambiguity”
Seção 3.2 e Tabela 12 ($r=-0.355$, $p<0.01$): “Role Ambiguity has a significant negative correlation with perceived work performance”.

− **Obi et al. (2025) — Identifying Factors Contributing to “Bad Days” for Software Developers: A Mixed-Methods Study**
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: “unclear project ownership”
Seção IV.A.1b (22 entrevistas): “there’s a remarkable amount of confusion about, like how work should be tracked”.

Desenho do trabalho

4 artigos · 0 positivas · 4 negativas

> Como a tarefa é montada: variedade de habilidade exigida, entregar algo inteiro ou só um pedaço sem sentido próprio, ter autonomia sobre o como, receber retorno do resultado. O caso ruim reconhecível é a fila de bug repetitivo, ou a tarefa mecânica atribuída sempre à mesma pessoa.

− **Ågren et al. (2022) — Agile software development one year into the COVID-19 pandemic**
Empirical Software Engineering · termo no artigo: “one reason is the lack of variation”
Seção 4.4.2 (entrevistado): “it is harder to keep up the motivation when working from home, compared to when working in the office, and one reason is the lack of variation”.

− **Graziotin et al. (2017) — On the unhappiness of software developers**
ACM International Conference Proceeding Series · termo no artigo: “Mundane or repetitive task”
Tabela 2 e Seção 4 (1.318 respondentes): causa nº 6 (60 menções) — ““Tedious”, “boring”, “dull”, “monotonous”, “trivial”, “recurrent”, etc., are the words the respondents used to describe the tasks that make them unhappy. “I tend to feel negative or bad when I am doing something that is not challenging, I instantly feel sleepy and bored””.

– **Massoni et al. (2019) — Relating voluntary turnover with job characteristics, satisfaction and work exhaustion-An initial study with Brazilian Developers**
Proceedings - 2019 IEEE/ACM 12th International Workshop on Cooperative and Human Aspects of Software Engineering, CHASE 2019 · termo no artigo: “most professionals lacked autonomy in the job they were willing to leave”
Seção IV.A (survey com 78 desenvolvedores que pediram demissão, JDS e Maslach): “Autonomy received the lowest scores regarding the developers’ last job, from which they left voluntarily. [...] inferior assessments were reported by most participants (52.6% of answers below 4). This result suggests most professionals lacked autonomy in the job they were willing to leave”; “On the other hand, at least two JCT core characteristics were positively assessed, on average: Job Significance and Skill Variety”.

– **Russo et al. (2023) — Satisfaction and performance of software developers during enforced work from home in the COVID-19 pandemic**
Empirical Software Engineering · termo no artigo: “lower when participants were bugfixing”
Seção 4.5 (estudo confirmatório, N=290; n=115 na atividade): “Activity satisfaction was on average lower when participants were bugfixing [M = -0.48, SD = 1.02, t(114) = -5.07, p< .0001]”; Tabela 7: competência também menor no bugfixing (M = −0.355, p = .0002).

Senso de realização

4 artigos · 3 positivas · 1 negativa

> Sensação de progresso: fechar tarefa, ver o código funcionando, entregar algo que serve. O contrário — travar num problema e passar o dia sem avançar — é um dos itens mais citados como o pior momento do dia, mais até que trabalho difícil ou volumoso.

+ **Madampe et al. (2023) — The Emotional Roller Coaster of Responding to Requirements Changes in Software Engineering**
IEEE Transactions on Software Engineering · termo no artigo: “the written codes work well”
Seção 4.3.2 (P65, Tester): “I felt all [high2 emotions] this when the written codes work well.” Estágio de testes: “Timely fixing of bugs and tests passing as expected trigger high1 emotions of the practitioners”.

+ **Santos et al. (2024) — Hybrid Work Well-Being: Software Professionals Finding Equilibrium**
IEEE Software · termo no artigo: “success in task completion”
Seção 4.1: “software professionals expressed their success in task completion within hybrid work environments. This sense of accomplishment often leads to feelings of fulfillment and satisfaction, making a substantial contribution to overall well-being”.

+ **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “task completion and delivery of results”
Seção 4.2.1: concluir tarefas e entregar resultados sustentou a competência, com orgulho relatado ao finalizar ajustes de interface (P5) — “Task completion and delivery of results, therefore, were other factors that affected participants’ competence”.

– **Graziotin et al. (2017) — On the unhappiness of software developers**
ACM International Conference Proceeding Series · termo no artigo: “Being stuck in problem solving”
Tabela 2 e Seção 4 (survey com 1.318 respostas completas de desenvolvedores, 219 causas de infelicidade): “Being stuck in problem solving is by far the most significant among the three factors” (186 menções, causa nº 1). Respondente: “I also thought of situations where I’m debugging some issue with the code and I can’t figure out why it isn’t working – when it seems like everything should work, but it just doesn’t. This is definitely one of the biggest gumption traps I encounter”.

Carga cognitiva

2 artigos · 0 positivas · 2 negativas

> Quanto a tarefa exige da cabeça ao mesmo tempo: contexto que precisa ser mantido, sistema que precisa ser entendido antes de mexer, decisões simultâneas. Diferente de volume — uma única tarefa num sistema emaranhado pode saturar mais que dez tarefas simples. É a razão pela qual pouca coisa na fila às vezes cansa mais que muita.

− **Brandebusemeyer et al. (2025) — Wearables to measure developer experience at work**
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "experienced cognitive load"
Seção IV.C (30 devs profissionais da SAP, 20 com dados fisiológicos): 66,7% relataram sinais de exaustão, e a exaustão correlaciona com a carga cognitiva percebida (τb=0.2501, p=0.0358), maior em tarefas de desenvolvimento (mediana 4.29 vs 3 em colaboração e 1.5 nas demais) — "high work exhaustion significantly correlates with high experienced cognitive load".

− **Meem e Johnson (2025) — Investigating the Impact of AI-Assisted Tools on Software Practitioner Well-Being**
CHIWORK 2025 - Adjunct Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work · termo no artigo: "cognitive overload"
Seção 4.1 (41 praticantes): "many respondents (65%) reported occasional feelings of being overwhelmed by the volume of information generated", e "Almost half of the participants (48%) reported experiencing mental fatigue".

Conflito de papéis

1 artigo · 0 positivas · 1 negativa

> Receber demandas incompatíveis entre si: dois gestores com prioridades opostas, ser dev e plantão de suporte no mesmo dia, cobrança por velocidade e por qualidade sem escolha declarada. Atender bem a um lado significa falhar no outro, e a falha é atribuída à pessoa.

− **Espina (2026) — Software Engineering Occupational Stress Factors and Work Performance**
ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management · termo no artigo: "Role Conflict"
Seção 3.2 (M=2.96): "software developers occasionally face overlapping responsibilities and competing expectations, especially in team-based or client-driven projects".

Técnico e ferramentas

9 fatores · 21 artigos distintos

Documentação interna

5 artigos · 4 positivas · 1 negativa

> Se dá para descobrir como as coisas funcionam sem interromper alguém: documentação que existe e está atualizada, decisão registrada com o motivo, informação que se acha. Onde falta, o trabalho simplesmente para até a pessoa certa responder — e o custo cai sobre quem foi interrompido também.

\+ **Madampe et al. (2024) — Supporting Emotional Intelligence, Productivity and Team Goals while Handling Software Requirements Changes**

ACM Transactions on Software Engineering and Methodology · termo no artigo: "tracking commitments and decisions"

Seção 4.6 (S9; P17): "Sometimes there's no note taker, or sometimes we are not actually recording it. So there are no meeting notes to refer back to. So a lot of times important information stages get forgotten after some point in time, but when you do it remotely, you have like an entire history of conversations that happen in the decisions".

\+ **Meyer et al. (2021) — Enabling good work habits in software developers through reflective goal-setting**

IEEE Transactions on Software Engineering · termo no artigo: "Keep Documentation Updated"

Seção 4.3 (8 desenvolvedores): "Developers also set goals to become more diligent at writing documentation and keeping it updated. With this goal, they not only aim to make their own work easier and more efficient, by remembering previous changes and decisions, but also to better support their co-workers".

\+ **Obi et al. (2025) — Identifying Factors Contributing to "Bad Days" for Software Developers: A Mixed-Methods Study**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "Documentation or Information Missing"

Seção IV.A (Fig. 4: 6,6% das respostas abertas): "when you can't find the knowledge you need to get your work done, that's very frustrating".

\+ **Vizcaíno et al. (2025) — Understanding Remote Work Experience: Insights Into Well-Being**

Journal of Software: Evolution and Process · termo no artigo: "Knowledge Sharing Problems"

Seção 4.4 (PLS-SEM, 184 respostas): "Hypothesis H4s is supported, as Knowledge Sharing Problems increase Stress (0.246). H4m indicates an inverse relationship with Motivation (−0.176)".

– **Micallef e Porter (2019) — Towards detecting and managing information anxiety in the ICT industry**

Proceedings of the International Conference on Software Engineering and Knowledge Engineering, SEKE · termo no artigo: "being unable to locate information and not having access to information"

Seção V.C (18 profissionais, coorte de QA): "the same cohort also has significant complaints of participants struggling with understanding, being unable to locate information and not having access to information". A definição do fenômeno é "stress caused by the inability to access, understand, or make use of information necessary for employees to do their job".

Ferramentas de IA para codificação

5 artigos · 4 positivas · 1 negativa

> Uso de assistente de IA para programar — Copilot e afins. Tira trabalho repetitivo, destrava começo de tarefa e reduz o custo de mexer no que não se conhece. Em contrapartida gera dependência, obriga a revisar código que ninguém escreveu e levanta dúvida sobre a própria habilidade estar enferrujando.

\+ **Kumar et al. (2025) — Time Warp: The Gap Between Developers' Ideal vs Actual Workweeks in an AI-Driven Era**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: “Frequency of AI Tool Usage”

Seção V.A (Fig. 7): “Developers who utilize AI tools on a daily basis experience the highest levels of satisfaction (74.5%)”, e “as AI usage frequency decreases, dissatisfaction becomes more prevalent”.

\+ **Upeksha et al. (2025) — The Influence of Generative AI on work-life balance among female software professionals in Sri Lanka**

2025 5th International Conference on Advanced Research in Computing: Converging Horizons: Uniting Disciplines in Computing Research through AI Innovation, ICARC 2025 - Proceedings · termo no artigo: “Generative AI”

Seção 4.2.4 (15 entrevistadas; R11): “Overall, Generative AI is seen as a positive influence on work-life balance. It helps reduce workload, saves time, and alleviates stress by offering quick solutions, enhancing code quality, and helping with complex concepts”; (R14): “It helps reduce cognitive load and provides clarity on complex tasks, which improves both productivity and job satisfaction”.

\+ **Valovy e Buchalcevova (2023) — The Psychological Effects of AI-Assisted Programming on Students and Professionals**

Proceedings - 2023 IEEE International Conference on Software Maintenance and Evolution, ICSME 2023 · termo no artigo: “Effectance”

Tabela I (amostra profissional, 5 desenvolvedores com experiência média de 11.2 meses de uso de ferramentas de IA). P2: "The AI tools save me time and prepare the code until I have to change some small parameters/nuances. They make me more effective and creative." P5 (Intrinsic Motivation): "It influenced my satisfaction levels positively as I do not have to worry about commenting, typing, or generating documentation." Polo negativo — P4: "I have become lazier and dependent. Before doing something, I prompt ChatGPT, then I take and edit it."

\+ **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**

Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “praised tools such as GitHub Copilot”

Seções 4.3.2 e 5.3: ferramentas de IA aumentaram eficiência e documentação, elevando o senso de competência (P2, P28) — “participants in our study praised tools such as GitHub Copilot, in which AI helped foster their competence through increased productivity”.

– **Meem e Johnson (2025) — Investigating the Impact of AI-Assisted Tools on Software Practitioner Well-Being**

CHIWORK 2025 - Adjunct Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work · termo no artigo: “AI-assisted tools”

Abstract e Seção 4.1 (survey preliminar, 41 praticantes): “while AI-assisted tools can increase productivity and improve focus, it can also cause mental fatigue, cognitive strain, and blurred work-life boundaries” — embora “80% of respondents reported that AI-assisted tools help them manage job-related stress”.

Manutenção do ambiente de desenvolvimento

5 artigos · 0 positivas · 5 negativas

> Atrito de máquina, ferramenta e infraestrutura: build que falha sem motivo aparente, ambiente que quebra depois de atualizar, pipeline lento, VPN caindo, teste instável. É tempo gasto sem produzir nada e sem aprender nada, o que o torna especialmente desmoralizante.

− **Kumar et al. (2025) — Time Warp: The Gap Between Developers' Ideal vs Actual Workweeks in an AI-Driven Era**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "maintaining development environment"

Seção IV.B (Tabela III: coef −0.0151, p=0.004) e D81: "Setting up the Dev. environment could be way simpler. Right now it is somewhat involved, time consuming, and brittle."

− **Obi et al. (2025) — Identifying Factors Contributing to "Bad Days" for Software Developers: A Mixed-Methods Study**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "Engineering System Friction"

Seção IV.A (22 entrevistas + survey com 214 devs; nº 1 da Tabela II, média 4.22; 14,7% das respostas abertas) (P04): "the build tools, the environment tools, ADO, Pull requests, Git, Visual Studio automation, every single tool that we use feels like it's barely working most days".

− **Ragkhitwetsagul et al. (2025) — The Impact of COVID-19 and Remote Work on Software Development in Thailand**

Proceedings - 9th International Conference on Information Technology, InCIT 2025 · termo no artigo: "Friction with collaboration tools"

Seção IV.E.2 (194 devs): "the top five most important challenges chosen by the developers are C8 Connectivity problems (75.00%), C17 Poor home work environment (74.47%), C14 Friction with collaboration tools (73.40%), C7 Insufficient hardware (71.43%)".

− **Santos et al. (2024) — Hybrid Work Well-Being: Software Professionals Finding Equilibrium**

IEEE Software · termo no artigo: "infrastructure-related problems"

Seção 4.2: "we have also documented reports from professionals who faced infrastructure-related problems, including issues with internet connectivity and the necessity for technical support to perform their tasks effectively. When infrastructure problems remain unaddressed, they have the potential to impact individual productivity, diminishing the sense of accomplishment and, in turn, affecting the overall well-being of software professionals."

− **Siitonen et al. (2025) — Coping with technostress in the software industry: Coping strategies and factors underlying their selection**

Journal of Systems and Software · termo no artigo: "unbearable IT tools"

Seção 4.1.1 (CIT, 715 relatos; escape = 227 menções): "one of the respondents described how they found the user experience of a specific debugger so unbearable that, in the end, they refused to use that tool again (IT avoidance). Others also decided to change jobs to work with tools and technologies that they found more enjoyable or at least less taxing".

Suporte e treinamento de TI

4 artigos · 4 positivas · 0 negativas

> Ter equipamento adequado, suporte que responde e treinamento na ferramenta que se é obrigado a usar. Inclui a ergonomia física do posto de trabalho, que em home office passou a ser responsabilidade de fato do próprio trabalhador — com resultado desigual.

\+ **Akdur et al. (2024) — Understanding Virtual Onboarding Dynamics and Developer Turnover Intention in the Era of Pandemic**

Journal of Systems and Software · termo no artigo: "Technology/Equipment delivery"

Abstract e Seção 5 (H2 suportada, B = 0.150): "This result emphasizes how important logistical support is to the effectiveness of remote work arrangements"; Apêndice A (TechD1): "Receiving work laptop, and equipment via mail/cargo was important, especially during the pandemic".

\+ **Ralph et al. (2020) — Pandemic programming: How COVID-19 affects software developers and how their organizations can help**

Empirical Software Engineering · termo no artigo: "Home office ergonomics"

Seção 5.5 (survey com 2225 respostas de 53 países; modelagem de equações estruturais, CFI = 0.961): "Based on this model, Hypotheses H1–H3, H5, H6, and H8–H10 are supported [...] change in wellbeing depends on ergonomics and fear". Abstract: "To improve employee productivity, software companies should focus on maximizing employee wellbeing and improving the ergonomics of employees' home offices".

\+ **Siitonen et al. (2025) — Coping with technostress in the software industry: Coping strategies and factors underlying their selection**

Journal of Systems and Software · termo no artigo: "IT support and training"

Seção 5.1: "the combination of limited organizational support for coping (e.g., training, control, and autonomy) and depleted personal resources can hinder the coping efforts of employees and lead to employees feeling helpless and exhausted".

\+ **Weerasekara et al. (2020) — ICT supported occupational stress management: The views of hr managers in Sri Lankan software companies**

Proceedings of the 12th IADIS International Conference e-Health 2020, EH 2020 - Part of the 14th Multi Conference on Computer Science and Information Systems, MCCSIS 2020 · termo no artigo: "Workplace Ergonomics"

Seção 5.1.1 (10 gerentes de RH de 8 empresas de software do Sri Lanka): "The participants strongly believed, since employees spend a lot of time inside the building, that the physical environment influences employee well-being and has direct impact on the employees' work performance and productivity". HR7: "There are pool tables and relaxation areas, the pantry area is also quite large so they can use it to gather and relax, the sick room is there".

Imposição de ferramentas

3 artigos · 0 positivas · 3 negativas

> Ter ferramenta, IDE, stack ou processo definido de cima, sem poder trocar o que comprovadamente atrapalha. O ponto não é a ferramenta ser boa ou ruim: é não ter escolha nem canal para mudar, o que transforma um problema técnico resolvível em desgaste recorrente.

– **Graziotin et al. (2017) — On the unhappiness of software developers**

ACM International Conference Proceeding Series · termo no artigo: "Imposed limitation on development"

Tabela 2 e Seção 4 (1.318 respondentes): causa nº 9 (40 menções) — "issues in the technical infrastructure a software project relies on often contribute to negative feelings among developers, especially when it is supposed to support software development, but instead imposes constraints or limitations".

– **Siitonen et al. (2025) — Coping with technostress in the software industry: Coping strategies and factors underlying their selection**
Journal of Systems and Software · termo no artigo: “Lack of control over IT”
Seções 4.1.2 e 4.1.4 (Tabela 4): “Those respondents who worked with, for example, client software or had strict limitations regarding the tools they were allowed to use, usually did not have the freedom to make these kinds of switches”; a falta de controle leva a “giving up”, “cognitive exhaustion and self-doubt”.

– **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “the leadership team dictated specific tool use”
Seção 4.3.2: a liderança ditar quais ferramentas usar frustrou a autonomia (P4, P31) — “the need for autonomy was unmet when the leadership team dictated specific tool use”.

Complexidade do código

2 artigos · 0 positivas · 2 negativas

> Quão emaranhado é o pedaço em que se mexe: muitas dependências, acoplamento alto, mudança pequena que estoura em cascata em lugares que ninguém previu. Trabalhar em componente complexo cansa mais e dá medo — e o medo, não a dificuldade, é o que faz a pessoa evitar a mudança necessária.

– **Madampe et al. (2023) — A Framework for Emotion-Oriented Requirements Change Handling in Agile Software Engineering**
IEEE Transactions on Software Engineering · termo no artigo: “complexity is high”
Seção V.B (surveys Alpha n=40 e Beta n=201): “when an RC is complicated to work on, they tend to feel more low1 emotions (e.g.: anxiety)”; “When the cascading impact of the RC is high, practitioners tend to feel both low2 and low1 emotions”. βP82: “Client requested a new functionality that would need to make a lot of changes in the project that was developed with a high level of complexity”.

– **Tulili et al. (2025) — Analyzing the Developer’s Sentiment in Software Components: A Decade-Long Study of the Apache Project**
Proceedings of the International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications · termo no artigo: “the complexity of software components”
Seção 6 (mineração do Apache httpd 2015–2024: 47.104 e-mails e 162.289 commits): “components with higher dependencies tend to generate more negative sentiments, suggesting that managing technical issues, such as file dependencies and complexity, is not just a matter of code but also of maintaining developer well-being”; Seção 4.2: “only ‘1:1’ is distinctly impacted by emotional expressions”.

Dívida técnica

2 artigos · 0 positivas · 2 negativas

> Código que todos sabem que está ruim e continua ali: gambiarra antiga, ausência de teste, dependência velha, parte que ninguém entende mais. Cada mudança custa mais tempo e mais medo, e conviver com isso todo dia é um dos itens que mais mobiliza frustração no trabalho técnico.

– **Graziotin et al. (2017) — On the unhappiness of software developers**
ACM International Conference Proceeding Series · termo no artigo: “Bad code quality and coding practice”
Tabela 2 e Seção 4 (1.318 respondentes): causa nº 3 (107 menções) — “It comes as no surprise that bad code quality and coding practices make developers unhappy. In almost all cases, bad code was a cause of unhappiness if it was written by other developers”. Também nº 7 (57 menções): “unexplained broken code: “When you haven’t changed the code, and suddenly the project doesn’t compile anymore. Worst feeling ever””.

– **Olsson et al. (2021) — Measuring affective states from technical debt: A psychoempirical software engineering experiment**

Empirical Software Engineering · termo no artigo: “technical debt activates a substantial portion of the emotional spectrum”

Abstract e Seção 6 (experimento de medidas repetidas com 40 profissionais de 12 empresas, n=200 observações, mais survey e entrevistas): “it is clear that technical debt activates a substantial portion of the emotional spectrum and is psychologically taxing”. Tema Apprehension (entrevistado): “The spontaneous feeling was a bit stress about too much stuff going on. Too many components, and some strange dependencies. And too much inheritance. [...] ’Cause I can see myself maintaining that code. And I can see that code breaking in the long term”. Na análise bayesiana, design smells impactam valência e dominância (Tabela 4).

Ergonomia visual

1 artigo · 1 positiva · 0 negativas

> Condições de passar o dia inteiro olhando tela: contraste, tema claro ou escuro, tamanho de fonte, iluminação da sala, quantidade e posição de monitor. É pouco discutido e tem efeito direto e mensurável em vista cansada, dor de cabeça e capacidade de sustentar foco à tarde.

\+ **Senyer (2026) — Circadian and color analysis of programming themes: effects of theme design, ambient lighting, and viewing distance on developer visual ergonomics**

Displays · termo no artigo: “developer visual ergonomics”

Seções 4.7 e 5 (análise espectrorradiométrica de 10 temas de IDE, sem participantes): “Such implementations address both immediate visual comfort and long-term developer well-being”.

Qualidade dos requisitos

1 artigo · 1 positiva · 0 negativas

> Se o que se pede está escrito de forma clara e suficiente para ser feito. Requisito impreciso não impede começar — o que ele faz é gerar discussão e retrabalho depois de pronto, quando a divergência de entendimento finalmente aparece.

\+ **Madampe et al. (2023) — A Framework for Emotion-Oriented Requirements Change Handling in Agile Software Engineering**

IEEE Transactions on Software Engineering · termo no artigo: “definition is imprecise or unclear”

Seção V.B (Definition): “When the RC is well-defined, practitioners tend to feel high1 emotions in general. [...] when RCs are not well-defined, practitioners tend to feel low2 emotions”. βP104: “I just hate projects where I need to run after information cause nobody knows anything, it tends to put me in a negative mood”.

Processo de desenvolvimento

5 fatores · 12 artigos distintos

Interação com clientes

8 artigos · 0 positivas · 8 negativas

> Lidar diretamente com cliente e com requisito que muda: mudança pedida na véspera do prazo, escopo entrando sem sair nada, cliente indisponível para tirar dúvida ou hostil na cobrança. O desgaste central é trabalho já feito virando trabalho perdido.

– **Brandebusemeyer et al. (2025) — Wearables to measure developer experience at work**
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "changing product requirements"
Seção IV.F (RQ4): mudanças de requisitos lideram o ranking de desafios que afetam os 30 participantes — "Changing product requirements impacts half of the participants to a great extent".

– **Espina (2026) — Software Engineering Occupational Stress Factors and Work Performance**
ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management · termo no artigo: "Client Interactions"
Seção 3.3: demanda moderada (M=3.02), dominada por mudanças frequentes de requisitos pelo cliente (item M=3.42) e requisitos incompletos (M=3.12) — "client-related stress arises mainly from changing project requirements and unclear specifications".

– **Madampe et al. (2023) — The Emotional Roller Coaster of Responding to Requirements Changes in Software Engineering**
IEEE Transactions on Software Engineering · termo no artigo: "Last minute RCs introduced closer to a deadline"
Abstract e Seção 4.4 (survey mundial, 201 praticantes): "Last minute RCs introduced closer to a deadline especially violate emotional well-being of practitioners". P138 (Developer): "A client wanted several significant changes made shortly before a deadline. I was very angry because they should've mentioned this earlier in development and having to make broad sweeping change so close to the deadline was very frustrating."

– **Madampe et al. (2024) — Supporting Emotional Intelligence, Productivity and Team Goals while Handling Software Requirements Changes**
ACM Transactions on Software Engineering and Methodology · termo no artigo: "Handling Software Requirements Changes"
Seção 4.5 (18 entrevistas + survey, 124 praticantes): "Software practitioners feel a wide range of emotions when handling RCs... The common emotions our participants mentioned that they feel during RC handling were excited, at ease, calm, content, relaxed, anxious, annoyed, frustrated, nervous, stressed, and depressed".

– **Madampe et al. (2024) — Addressing Bad Feelings in Agile Software Project Contexts: Considering Team Welfare and Developer Mental Health**
IEEE Software · termo no artigo: "unstable requirements"
Seção "Developer productivity and their emotions" (survey com 201 praticantes, escala JAWS): "Unstable requirements result in anxiety at the commencement of the project: 'I was anxious when we first started and the requirements were still not settled on.' – Tester"; "RCs introduced after design and implementation tends to make practitioners feel negative emotions".

– **Ribeiro (2022) — Understanding the relationships between the perceptions of burnout and instability in Software Engineering**
ACM International Conference Proceeding Series · termo no artigo: "instabilidade da tarefa"
Seção 2.2 (411 respondentes): "a instabilidade da tarefa é o resultado da introdução de novos requisitos, de repriorizações, e requisições ex-post. Ela é definida neste trabalho como a percepção do indivíduo sobre as mudanças que ocorrem em suas tarefas"; "a instabilidade da tarefa é a que os indivíduos percebem com maior frequência no desenvolvimento de software".

– **Romano et al. (2024) — MOOD: Mindfulness fOr sOftware Developers**
International Symposium on Empirical Software Engineering and Measurement · termo no artigo: “interacting with customers”
Seção 5 (3 grupos focais): “These interactions lead software developers to cope with some sources of stress, such as external pressures and expectations, time management scheduled by others, and understanding requests that are unclear, incomplete, or do not fit the context”.

– **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “changes made by stakeholders that resulted in wasted time”
Seção 4.1.1 (P10): “we will get done with a story [i.e., task or part of a feature] or will be building a story, and two weeks later will rip out that feature”.

Adoção de métodos ágeis

1 artigo · 0 positivas · 1 negativa

> Quanto e como o time realmente usa práticas ágeis: sprint, daily, revisão, XP. Puxa nos dois sentidos. Ritmo previsível e trabalho fatiado ajudam; cerimônia demais, daily que virou prestação de contas e sprint sem folga entre uma e outra levam a exaustão. O que decide o sinal é a dose e o motivo de cada prática, não a adoção em si.

– **Newman et al. (2025) — Disclosure of Neurodivergence in Software Workplaces: a Mixed Methods Study of Forum and Survey Perspectives**
ASSETS 2025 - Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility · termo no artigo: “standup”
Seção 5.4 (survey, AuDHD vs NT): “AuDHD participants were more likely to say that standup (a meeting/Agile practice common to software workplaces [117]) hindered their well-being, whereas NT participants were more likely to say that it helped ($\chi 2$, $p = .002$)”.

Alinhamento das práticas ágeis

1 artigo · 1 positiva · 0 negativas

> A distância entre o processo que o time pratica e o que cada pessoa precisa para trabalhar bem. Não é sobre ser ágil ou não: é sobre encaixe. Alguém que precisa de blocos longos de foco num time com três reuniões diárias sofre mesmo que o processo esteja formalmente correto. O desconforto costuma ser lido como resistência quando é incompatibilidade.

\+ **Benlian et al. (2025) — Team-Enacted Use vs. Developer-Needed Use of Agile Practices: How Perceptual (In-)Congruence and Team Feedback-Seeking Shape Developer Well-Being**
Information Systems Research · termo no artigo: “Team-Enacted Use versus Developer-Needed Use of Agile Practices”
Experience-sampling (149 desenvolvedores, 1.510 respostas diárias) e 32 entrevistas: “perceived congruence (versus incongruence) consistently leads to higher levels of developer well-being”.

Estruturação do processo de trabalho

1 artigo · 1 positiva · 0 negativas

> *Sem descrição no catálogo (pendência da tarefa 1.3).*

\+ **Stray et al. (2022) — Exploring human factors of the agile software tester**
Software Quality Journal · termo no artigo: “thrived in organized job environments”
Seção 4.4 (Informante 18): “I applied for a job and got it. And what do I encounter? No structure. […] I can’t work in these conditions, so I applied for another job”.

Gamificação no trabalho

1 artigo · 0 positivas · 1 negativa

> Pontuação, ranking, badge e meta com formato de jogo aplicados ao trabalho. Engaja no começo, mas desloca a atenção para o indicador: otimiza-se o que é contado, e o que não é contado deixa de ser feito. Também prende em perseguição de número que não importa.

– **Elsalmy et al. (2025) — Exploring Gamification Risks and Implications in Agile Software Development Enterprises: An Empirical Study**

ICSIE 2024 - 2024 13th International Conference on Software and Information Engineering · termo no artigo: “gamification risks”

Seções 4.1–4.5 (entrevistas com times scrum de 3 empresas + focus group com especialistas): pontos, badges, leaderboards e Easter eggs geraram medo de incompetência, procrastinação e vício, desmotivação, rejeição por pares e disrupção do fluxo — “I get so immersed into the system, and I keep looking for everyone’s achieved badges and Easter eggs [...] I leave important tasks unachieved” (P12).

Equipe e relações

10 fatores · 36 artigos distintos

Relações no trabalho

25 artigos · 23 positivas · 2 negativas

> Qualidade do convívio no time: confiança, ajuda mútua, conversa informal, conflito que se resolve, comunicação aberta. É o fator com mais evidência acumulada em todo o levantamento, e o mais afetado pelo trabalho remoto — o que se perdeu ali foi a interação não planejada, não a reunião.

+ **Akdur et al. (2024) — Understanding Virtual Onboarding Dynamics and Developer Turnover Intention in the Era of Pandemic**
Journal of Systems and Software · termo no artigo: "Workplace Relationship Quality"
Seção 5 (H6): "Workplace Relationship Quality (WRQ) also had a negative impact on Turnover Intention (TI) (B = −0.187, T = 2.675, p < 0.05)"; Seção 2.4: "high-quality workplace relationships directly impact employee engagement and well-being in virtual environments".

+ **Arriel et al. (2026) — Understanding developer well-being: measuring mental health and productivity in software teams**
Empirical Software Engineering · termo no artigo: "workplace relationships and social support"
Seção 6.1.3: ambientes comunicativos, compreensivos e com boas relações facilitaram manter a saúde mental (P7, P10) — "several responses underscored the importance of workplace relationships and social support in maintaining both mental health and productivity".

+ **Chattopadhyay et al. (2021) — Developers Who Vlog: Dismantling Stereotypes through Community and Identity**
Proceedings of the ACM on Human-Computer Interaction · termo no artigo: "the work culture of some organizations can be very supportive thus reducing stress"
Seção 5.2.5: "At the same time, vlogs also described how the work culture of some organizations can be very supportive thus reducing stress: "if you put your hand up the help will be there!" [V31]".

+ **Espina (2026) — Software Engineering Occupational Stress Factors and Work Performance**
ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management · termo no artigo: "Team Factors"
Seções 3.3 e 3.5: "Interpersonal and client-related stressors were lowest, particularly Team Factors (M = 2.38), indicating strong collaboration and team support".

+ **Hasan et al. (2021) — A Survey-based qualitative study to characterize expectations of software developers from five stakeholders**
International Symposium on Empirical Software Engineering and Measurement · termo no artigo: "Workplace Culture"
Seção 4.1.2 (survey com 181 desenvolvedores profissionais; expectativa mais citada em relação às organizações): "Participants expect their organizations to focus on workplace culture (48.78%) that covers overtime-free and politics-free environment, work-life balance, good relations, etc. R2,24 mentions "A friendly work environment with the best practice of modern methodologies"".

+ **Juarez-Ramirez et al. (2022) — Assessing Wellbeing and Soft Skills of Software Developers and Students in the Context of Working from Home during the COVID-19 era**
Proceedings - 2022 10th International Conference in Software Engineering Research and Innovation, CONISOFT 2022 · termo no artigo: "social isolation"
Seção V.A (survey com desenvolvedores da indústria e estudantes; dificuldades no trabalho remoto): "In the case of social isolation, significant percentages revealed this difficulty; developers (56%), students (58%)"; "Team interaction presents a significant difference, developers (47%), students (75%). Social experience is also significantly lived, developers (56%), students (67%). Again, social interaction is a preoccupation for both groups".

+ **Kuutila et al. (2021) — Individual differences limit predicting well-being and productivity using software repositories: a longitudinal industrial study**
Empirical Software Engineering · termo no artigo: "you would seek support with"
Seção 4.4 (Developer 1, sobre o uso de emoji no chat do projeto): "in those moments when you felt frustrated or irritated, then you would seek support with "in the trenches"-kind of humor".

+ **Madampe et al. (2023) — The Emotional Roller Coaster of Responding to Requirements Changes in Software Engineering**
IEEE Transactions on Software Engineering · termo no artigo: "positive team dynamics"
Seção 4.3.2 (estágio de desenvolvimento): "working as a team which is ready to embrace change, overcome difficulties, and be able to solve issues rapidly makes practitioners feel ecstatic and inspired". Polo negativo (P181): "I felt this emotion at the beginning because there were some obstacles with disagreements between the developers".

+ **Madampe et al. (2024) — Supporting Emotional Intelligence, Productivity and Team Goals while Handling Software Requirements Changes**
ACM Transactions on Software Engineering and Methodology · termo no artigo: "Open and Regular Communication"
Seção 4.6 (S2, 16 de 18 participantes; P16): "Perhaps the problems come in when you have more layers between managers and software developers and the customer... And that's when you get anger and frustration because that's when you start going to like, we're not working together anymore".

+ **Meyer et al. (2021) — Enabling good work habits in software developers through reflective goal-setting**
IEEE Transactions on Software Engineering · termo no artigo: "take short social breaks"
Seção 4.4 (At Work: Stay Motivated and Fresh; 11 desenvolvedores): "At work, developers want to have more regular and productive work hours to stay motivated and fresh. To achieve their goal, they regularly take short social breaks (S23) and use these breaks to have meaningful conversations with co-workers or learn from them".

+ **Neumann et al. (2022) — The Covid 19 Pandemic and its Effects on Agile Software Development**
ACM International Conference Proceeding Series · termo no artigo: "significant decrease in social exchange"
Seção 4.2 (estudo de caso múltiplo com 19 entrevistados em 3 casos): "The finding F3 can also be confirmed, as we notice a significant decrease in social exchange. By this, we understand all types of communication and collaboration that do not primarily have a professional context but promote social interaction"; "many interviewees point out that these events cannot replace personal, social exchange in the office". Seção 4.3: "Multiple interviewees from all of our cases tell that they miss the social exchange with their colleagues. [...] Presumably, this could lead to unstable teams who can no longer collaborate within their team".

+ **Obi et al. (2025) — Identifying Factors Contributing to "Bad Days" for Software Developers: A Mixed-Methods Study**
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "Issues around Team dynamics and communication"
Seção IV.A.1c (22 entrevistas; Tabela II média 4.11 para falta de suporte): "Developers often complained about difficulties in collaboration, communication breakdowns, unresponsive team members, and interpersonal conflicts".

+ **Ragkhitwetsagul et al. (2025) — The Impact of COVID-19 and Remote Work on Software Development in Thailand**
Proceedings - 9th International Conference on Information Technology, InCIT 2025 · termo no artigo: "Missing social interactions"
Seção IV.E.2 (194 devs): "the top five challenges include C1 Missing social interactions (80%), C6 Difficult to communicate with colleagues (71.05%), C2 Lack of work-life boundary (70.00%), C4 Less awareness of colleague work (67.89%)".

+ **Russo et al. (2023) — Satisfaction and performance of software developers during enforced work from home in the COVID-19 pandemic**
Empirical Software Engineering · termo no artigo: "disagreements or conflicts with people working on the same task"
Seção 4.3 (modelos lineares mistos, N=290): "Across all activities, activity satisfaction was negatively predicted by conflicts and pressure, and positively by autonomy, competence, and relatedness". Item da medida: "I had disagreements or conflicts with people working on the same task as me".

+ **Russo et al. (2024) — Understanding Developers Well-Being and Productivity: A 2-year Longitudinal Analysis during the COVID-19 Pandemic**

ACM Transactions on Software Engineering and Methodology · termo no artigo: “communication with colleagues”

Seção 4.3 (regressão com variáveis da Wave 6): “Of the 14 predictors, only solidarity, B = .18, SE = .05, p < .001, distractions at home, B = −.17, SE = .08, p = .04, need for relatedness, B = −.35, SE = .14, p = .01, and communication with colleagues, B = .31, SE = .08, p < .001 were associated with job satisfaction”. Seção 5.2: “the quality of social contacts correlated positively with well-being and improved between waves 1 and 6”.

+ **Santos et al. (2024) — Hybrid Work Well-Being: Software Professionals Finding Equilibrium**

IEEE Software · termo no artigo: “reduced interaction with teammates and co-workers”

Seção 4.2: “software professionals may experience increased family time but reduced interaction with teammates and co-workers. [...] These workplace relationships foster a sense of belonging, mutual support, job satisfaction, motivation, and effective teamwork and collaboration”.

+ **Siitonen et al. (2025) — Coping with technostress in the software industry: Coping strategies and factors underlying their selection**

Journal of Systems and Software · termo no artigo: “seeking social contact in the workplace”

Seção 4.2.1: “many of those struggling with technical problems or troublesome development tools felt like other developers would be the most likely to understand and accept their struggles and encourage them to keep pushing forward toward a solution”.

+ **Thasipan et al. (2024) — Determinants of Employee Engagement in the Post-Pandemic Working-from-Home Contexts: Evidence from the Sri Lankan Software Industry**

Proceedings - International Research Conference on Smart Computing and Systems Engineering, SCSE 2024 · termo no artigo: “Social interaction”

Seção V.E (H4 aceita, coeficiente 0.468): “The positive impact of increased social interaction on employee engagement emphasizes recreating social dynamics in virtual settings”. Sobre a colaboração em pequenos grupos (H5, mediação parcial): “The partial mediating effect of small-group collaboration emphasizes the importance of fostering collaborative spaces in virtual environments”.

+ **Trinkenreich et al. (2025) — Guidelines for Cultivating a Sense of Belonging to Reduce Developer Burnout**

IEEE Software · termo no artigo: “spontaneous office exchanges and feedback loops”

Seção Create Optional In-Person Opportunities to Meet: “Encountering one another, even if not on a daily basis and particularly without the compulsion of an on-site work arrangement, contributes to cultivating a sense of belonging and alleviates job burnout”.

+ **Valovy e Buchalcevova (2023) — The Psychological Effects of AI-Assisted Programming on Students and Professionals**

Proceedings - 2023 IEEE International Conference on Software Maintenance and Evolution, ICSME 2023 · termo no artigo: “Relaxed dynamics with AI”

Tabela I (tema Dynamics of Human and AI Pairing; amostra profissional). P5: "With a human, it (pairing) is more energetic and stressful and can lead to better results than with an AI tool. With AI, it is more relaxed." P4: "I am quite an introverted person so definitely prefer the AI tool. With a human, I am always a little nervous or trying to show off how I can do things." P1: "AI does not match up to a human collaborator."

+ **Vizcaíno et al. (2025) — Understanding Remote Work Experience: Insights Into Well-Being**

Journal of Software: Evolution and Process · termo no artigo: “trust among team members”

Seção 4.4 (PLS-SEM, 184 respostas): “Trust weakly reduces developers' Stress (−0.115) while increasing their Motivation (0.228)”; “Communication reduces Stress (−0.142)... and H3m is supported, as the value indicates that Communication increases Motivation (0.230)”.

+ **Weerasekara et al. (2020) — ICT supported occupational stress management: The views of hr managers in Sri Lankan software companies**
Proceedings of the 12th IADIS International Conference e-Health 2020, EH 2020 - Part of the 14th Multi Conference on Computer Science and Information Systems, MCCSIS 2020 · termo no artigo: "Cohesive Team and Teamwork"
Seção 5.1.3: "A shared practice identified in all the companies was the "teamwork." [...] The respondents figured that the team members were more open to the supervisor and their peers, and that such structure tended to alleviate communication regarding work as well as obstacles they were facing". HR2: "We practice teamwork, employees recognize (company name) as a family".

+ **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "find social support"
Seção 4.1.2 (P4): "go out and socialize with coworkers over lunch time, which also again also helps you decompress let off some steam"; (P7): "I need people around me. I may not talk to them. But I would rather [be] in office, and knowing that people [are] around". Em contraste no remoto (P8): "every single interaction, more or less, it's almost like it needs to be planned".

– **Ågren et al. (2022) — Agile software development one year into the COVID-19 pandemic**
Empirical Software Engineering · termo no artigo: "The lack of interaction may affect peoples' feelings and personalities"
Seção 4.4.1 (7 entrevistados): "The lack of interaction may affect peoples' feelings and personalities, as well as the ability to gain advantages of working as a team".

– **Graziotin et al. (2017) — On the unhappiness of software developers**
ACM International Conference Proceeding Series · termo no artigo: "Under-performing colleague"
Tabela 2 e Seção 4 (1.318 respondentes): causa nº 4 (71 menções) — "the under-performance of colleagues, either team members, colleagues in other teams, or external collaborators, most often make developers experience negative feelings". Respondente: "Last time I felt angry when a senior developer again committed an update ruining a beautiful generic solution I've made before".

Segurança psicológica

9 artigos · 8 positivas · 1 negativa

> Poder dizer "não sei", "errei" ou "isso não vai dar certo" sem custo pessoal. Onde falta, o problema aparece tarde, a estimativa é inflada por precaução e ninguém pede ajuda antes de já ter perdido dias. Inclui clareza sobre as normas do time e aceitação de que estresse existe.

+ **Cerqueira et al. (2024) — Empathy and Its Effects on Software Practitioners' Well-Being and Mental Health**
IEEE Software · termo no artigo: "a culture where failure is embraced"
Seção Being Open (Developer Advocate) e Takeaways: "Cultivating a culture where failure is embraced, and a culture of experimentation is cultivated is important and essential"; "Fostering an empathetic workplace culture where team members can trust, admit mistakes, and accept failures can reduce blame, improve job motivation, and build stronger human connections".

+ **Gunatilake et al. (2026) — Empathy Guidelines for Improving Practitioner Well-Being and Software Engineering Practices**
IEEE Software · termo no artigo: "Creating a safe space"
Tabela 1 (G10), quadrante Quick Wins: "Fostering a safe environment where team members can share concerns without fear of judgement promotes well-being and positive team dynamics".

+ **Kennedy et al. (2026) — Agile Practices and IT Development Team Well-Being: Unveiling the Path to Successful Project Delivery**
Engineering Management Journal · termo no artigo: "psychological safety positively affects job satisfaction in agile projects"
Seção Hypothesis Test Results (survey com equipes ágeis): "Research shows that psychological safety positively affects job satisfaction in agile projects (H9) ($\beta = 0.91$, $t = 25.559$, $p < .001$)".

+ **Madampe et al. (2024) — Supporting Emotional Intelligence, Productivity and Team Goals while Handling Software Requirements Changes**
ACM Transactions on Software Engineering and Methodology · termo no artigo: “degree of psychological safety”
Seção 4.6 (S1): “in large groups or when their degree of psychological safety is low, team members are unwilling to express their ‘raw’ emotions in front of others in the team. In contrast, they said that small groups better promote the freedom to express these emotions”.

+ **Siitonen et al. (2025) — Coping with technostress in the software industry: Coping strategies and factors underlying their selection**
Journal of Systems and Software · termo no artigo: “Accepting organizational culture regarding (techno)stress”
Seção 4.2.1 (74 respondentes): “An accepting organizational culture regarding stress was also a relevant factor here because it made it easier to talk about one’s burden and seek peer support and understanding”.

+ **Sánchez-Gordón et al. (2026) — The role of voice and silence behaviors in software development: a structural equation modeling analysis**
Information and Software Technology · termo no artigo: “Psychological safety”
Tabela de resultados do SEM (158 respostas válidas de profissionais de desenvolvimento de software), H1 aceita: “Psychological safety is inversely related to both the software developer’s silence (H1) and voice (H2), but it is significantly and more strongly associated with changes in his/her silence”. H2 (segurança psicológica sobre voz) foi rejeitada.

+ **Trinkenreich et al. (2024) — Unraveling the Drivers of Sense of Belonging in Software Delivery Teams: Insights from a Large-Scale Survey**
Proceedings - International Conference on Software Engineering · termo no artigo: “Psychological Safety”
Seção 6.1.2 (H2 aceita): “We found support for the link between psychological safety and belonging to the team (H2, B=.410)”. Resposta à RQ1: “Notably, psychological safety exhibits a stronger correlation with the sense of belonging compared to work appreciation”.

+ **Trinkenreich et al. (2025) — Guidelines for Cultivating a Sense of Belonging to Reduce Developer Burnout**
IEEE Software · termo no artigo: “Psychological safety”
Tabela 1 (Estudo 3, 10.781 respostas; ICSE 2024): “Psychological safety and work appreciation positively correlate with a sense of belonging to the team”; seção Conduct Blameless Postmortems: “This kind of feedback improves psychological safety and builds trust across the team”.

– **Ågren et al. (2022) — Agile software development one year into the COVID-19 pandemic**
Empirical Software Engineering · termo no artigo: “some people feel less safe in online meetings”
Seção 4.4.2 (7 entrevistados): “it became clear that some people feel less safe in online meetings, and that without psychological safety, some discussions will not be raised”.

Empatia no trabalho

4 artigos · 4 positivas · 0 negativas

> Colegas e gestor levarem em conta a situação concreta do outro ao cobrar, revisar e distribuir tarefa. Aparece em coisas específicas: como um feedback difícil é dado, se alguém pergunta antes de assumir má intenção, se a pessoa em dificuldade é ajudada ou marcada.

+ **Cerqueira et al. (2024) — Empathy and Its Effects on Software Practitioners' Well-Being and Mental Health**
IEEE Software · termo no artigo: “practicing empathy”
Seção Practicing Empathy (síntese temática de 22 artigos de praticantes da comunidade DEV): “By practicing empathy, software practitioners can increase their well-being and mental health as their coworkers will consider their emotions, feelings, and perspectives”; Abstract: “practice empathy by being mindful, being open, understanding others, and taking care, which can reduce blame, improve job motivation, prevent burnout, and create a better work environment”.

+ **Gunatilake et al. (2026) — Empathy Guidelines for Improving Practitioner Well-Being and Software Engineering Practices**
IEEE Software · termo no artigo: "embedding empathy into daily practice"
Seções Perceived Importance e Adoption Willingness (survey com 103 praticantes): "Several practitioners also highlighted empathy's role in reducing stress, promoting well-being, and aligning team goals".

+ **Madampe et al. (2023) — A Framework for Emotion-Oriented Requirements Change Handling in Agile Software Engineering**
IEEE Transactions on Software Engineering · termo no artigo: "A manager having empathy"
Seção V.C (RQ3.3): "when a manager does not pressurise their team, but motivates the team (MC1) by being empathetic and allowing the team to handle their RCs as best suits them, practitioners tend to experience an arising of high2 emotions (e.g.: energetic), and high1 (e.g.: calm) emotions". Polo negativo (βP45): "my ex team leader was suddenly angry for every minor mistake".

+ **Madampe et al. (2024) — Supporting Emotional Intelligence, Productivity and Team Goals while Handling Software Requirements Changes**
ACM Transactions on Software Engineering and Methodology · termo no artigo: "Empathising"
Conclusão e Seção 4.6 (S4): "Empathising with others and tracking commitments and decisions as a team are key strategies that have strong correlations between managing emotions, between sustaining team productivity, and between setting and sustaining team goals"; (P14): "most of it starts with just trying to validate and listen to the person and why they have the concern".

Integração de novatos

3 artigos · 3 positivas · 0 negativas

> Como quem chega é recebido nas primeiras semanas: acesso funcionando no primeiro dia, alguém explicitamente responsável por acompanhar, primeira tarefa possível de concluir, apresentação ao time. Onboarding remoto malfeito é um dos motivos mais citados de saída precoce.

+ **Akdur et al. (2024) — Understanding Virtual Onboarding Dynamics and Developer Turnover Intention in the Era of Pandemic**
Journal of Systems and Software · termo no artigo: "Virtual onboarding success"
Seção 5 (PLS-SEM, 150 devs Salesforce Commerce Cloud contratados na pandemia): "we found that VOS had a substantial and positive effect on job satisfaction (JS) ($B = 0.717$, $T = 17.459$, $p < 0.001$)"; "VOS significantly and positively affected the quality of the work-life relationship (WRQ) ($B = 0.566$, $T = 9.514$, $p < 0.001$)".

+ **Neumann et al. (2022) — The Covid 19 Pandemic and its Effects on Agile Software Development**
ACM International Conference Proceeding Series · termo no artigo: "When new employees are integrated into a team"
Seção 4.3: "When new employees are integrated into a team, the effects are noticeable. This is perceived as more challenging and complex, especially in pure remote working models. The duration of induction, socialization in the team, and the time, until new employees perform are perceived as more extended than in the co-located working model".

+ **Weerasekara et al. (2020) — ICT supported occupational stress management: The views of hr managers in Sri Lankan software companies**
Proceedings of the 12th IADIS International Conference e-Health 2020, EH 2020 - Part of the 14th Multi Conference on Computer Science and Information Systems, MCCSIS 2020 · termo no artigo: "Buddy"
Seção 5.1.3: "The "Buddy" concept had been introduced in some companies, along with the team concept. The buddy was described as a peer working in the same project and assigned to co-work with new employees. The buddy was responsible for and supported the work assignments, and this person helped others to adjust to the work setting and work culture".

Mentoria

3 artigos · 3 positivas · 0 negativas

> Ter alguém mais experiente disponível, com ou sem formalidade: pair programming, revisão que explica em vez de só apontar, conversa de carreira. Vale nas duas pontas — quem ensina também relata ganho, e a relação próxima entre quem orienta e quem é orientado é o que faz diferença, mais que o programa.

\+ **Cardenas-Castro et al. (2019) — Soft skills training: Performance psychology applied to software development**

Proceedings - 2019 IEEE/ACM 12th International Workshop on Cooperative and Human Aspects of Software Engineering, CHASE 2019 · termo no artigo: "having a close relationship between the trainer and the trainee"

Seção IV: "This experience allowed the psychology team to establish the importance of having a close relationship between the trainer and the trainee, giving to the later an opportunity to receive valuable personal information for the training process faster. Also, a close relationship allows the trainee to develop trust in the accompaniment process and enhances mutual feedback".

\+ **Russo et al. (2023) — Satisfaction and performance of software developers during enforced work from home in the COVID-19 pandemic**

Empirical Software Engineering · termo no artigo: "helping others"

Seção 4.5 (estudo confirmatório, N=290): "Activity satisfaction was on average lower when participants were bugfixing [M = -0.48, SD = 1.02, t(114) = -5.07, p< .0001], and higher when participants were helping others [M = 0.56, SD = 0.77, t(35) = 4.39, p = .0001]"; "Competence was lower when bugfixing and higher when helping people. Relatedness was only higher when people were helping."

\+ **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**

Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "formal or informal mentorship"

Seções 4.1.2 e 4.2.2: mentoria técnica e interpessoal ajudou a ganhar competência, e sua falta após reorganizações deixou desenvolvedores sem suporte (P21, P24, P27) — "having technical and soft skill-based mentorship helped them gain the skills needed to feel competent".

Competição no trabalho

2 artigos · 0 positivas · 2 negativas

> Quando o ambiente coloca colega contra colega: ranking de produtividade, bônus por métrica individual, comparação pública de entrega. Mata a ajuda mútua justamente onde ela é mais barata — ninguém para para desbloquear o outro se isso piora o próprio número.

– **Arriel et al. (2026) — Understanding developer well-being: measuring mental health and productivity in software teams**

Empirical Software Engineering · termo no artigo: "Unhealthy competition"

Seção 6.1.3 (P13, P14, P16, P21): "reward-based productivity structures contribute to stress, anxiety, and unhealthy competition".

– **Elsalmy et al. (2025) — Exploring Gamification Risks and Implications in Agile Software Development Enterprises: An Empirical Study**

ICSIE 2024 - 2024 13th International Conference on Software and Information Engineering · termo no artigo: "Social comparison & competitive pressure"

Seção 4.4 (P6, P8): "Competition [Game element: Competitions] even in a training shakes my confidence, it affects me mentally".

Diversidade de competências

2 artigos · 2 positivas · 0 negativas

> Time com repertório variado — funções, trajetórias e experiências diferentes — e, sobretudo, ter perto de si alguém que sabe o que você não sabe. O efeito mais imediato é reduzir tempo travado: a dúvida encontra resposta dentro do time em vez de virar dias de tentativa.

\+ **Madampe et al. (2023) — A Framework for Emotion-Oriented Requirements Change Handling in Agile Software Engineering**

IEEE Transactions on Software Engineering · termo no artigo: "peers with the necessary knowledge"

Seção V.C (RQ3.2): "Having team members who are experts, competent, and having relevant experience in the team tend to arise high2 (e.g.: inspired) and high1 (e.g.: relaxed, satisfied, calm, at ease) emotions in practitioners when they are handling RCs". βP107: "I belonged to an experienced team. [...] We had every reason to stay calm and implement changes in the same way we had done multiple times before".

\+ **Vizcaíno et al. (2025) — Understanding Remote Work Experience: Insights Into Well-Being**

Journal of Software: Evolution and Process · termo no artigo: "Skills and Experience"

Seção 4.4 (PLS-SEM, 184 respostas válidas): "Hypothesis H1s and H1m are supported, indicating that Skills and Experience reduce developers' Stress (−0.225) while simultaneously increasing their Motivation (0.313)".

Comunicação de más notícias

1 artigo · 0 positivas · 1 negativa

> *Sem descrição no catálogo (pendência da tarefa 1.3).*

– **Stray et al. (2022) — Exploring human factors of the agile software tester**

Software Quality Journal · termo no artigo: "the testers were often bearers of bad news"

Seção 4.3 (Informante 15): "You have to talk to people, and not everyone is super-pleased to hear what you have to say. People are stressed and not really glad to hear about problems".

Fadiga por empatia

1 artigo · 0 positivas · 1 negativa

> Desgaste de quem sustenta o emocional dos outros: tech lead que virou apoio informal do time inteiro, pessoa sênior que todos procuram para desabafar. Cuidar também esgota, e costuma ser invisível porque a pessoa nessa posição é justamente a que parece estar bem.

– **Gunatilake et al. (2026) — Empathy Guidelines for Improving Practitioner Well-Being and Software Engineering Practices**

IEEE Software · termo no artigo: "empathy fatigue"

Seção Perceived Importance e Tabela 1 (G17): "Excessive empathy can cause fatigue, so setting boundaries and prioritising self-care is essential".

Instabilidade da equipe

1 artigo · 0 positivas · 1 negativa

> Time cuja composição muda com frequência, por saída, entrada ou realocação. Cada troca reinicia contexto, confiança e combinado tácito, e o custo recai sobre quem permanece — que refaz onboarding e reexplica decisão a cada rodada.

− **Ribeiro (2022) — Understanding the relationships between the perceptions of burnout and instability in Software Engineering**

ACM International Conference Proceeding Series · termo no artigo: "instabilidade da equipe"

Seções 2.2 e 6 (survey transversal com 411 respondentes do processo de desenvolvimento de software, análise fatorial confirmatória): "a percepção da instabilidade da equipe é definida como a percepção dos indivíduos quanto às mudanças que ocorrem em relação à saída e à entrada de membros que influenciam as atividades da equipe". Resultado: "A relação entre a percepção do burnout e a instabilidade da equipe, tecnológica e da tarefa, é positiva e significativa para as dimensões da exaustão e do cinismo".

Organizacional

20 fatores · 29 artigos distintos

Apoio do gestor

12 artigos · 11 positivas · 1 negativa

> O que o gestor imediato faz na prática: dá contexto, remove obstáculo, protege de demanda absurda, redistribui carga, dá retorno de verdade e conversa a sério num a um — ou micromanageia, decide sem envolver e desaparece quando precisa. É um dos fatores com mais peso, e costuma superar política de empresa: a experiência de trabalho de alguém é em boa medida a experiência de ter aquele gestor.

\+ **Brandebusemeyer et al. (2025) — Wearables to measure developer experience at work**
IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "Better support / more feedback"
Seção IV.F e Tabela III: "Better support / more feedback".

\+ **Chilton et al. (2010) — Performance and Strain Levels of IT Workers Engaged in Rapidly Changing Environments: A Person-Job Fit Perspective**
Data Base for Advances in Information Systems · termo no artigo: "needs for supervisory support"
Seção Building the Hypotheses (definição operacional): "we limit our discussion here to the degree to which a supervisor becomes involved in a worker's professional development by providing the tools and training needed to accomplish a job; by providing advice, feedback and credit during and after the work". Exemplo do mecanismo: "When this individual moves to a new job environment in which the perceived level of supervisory support drops, a deficit occurs such that needs exceed supplies. As a result of this decrease in fit, the individual's work processes may be less efficient and work outcomes may be of a lower quality".

\+ **Gunatilake et al. (2026) — Empathy Guidelines for Improving Practitioner Well-Being and Software Engineering Practices**
IEEE Software · termo no artigo: "Flexibility in handling human issues"
Tabela 1 (G12): falta de empatia em crises pessoais levou a pedidos de demissão, enquanto apoio empático gerou lealdade renovada — "empathetic support allowed team members to handle emergencies and return with renewed loyalty".

\+ **Guru et al. (2025) — Cultivating Agile Software Development Teams through IT Servant Leadership: Strategies for Success**
EECSI 2025 - Proceedings 2025 12th International Conference on Electrical Engineering, Computer Science and Informatics · termo no artigo: "implementing servant leadership practices"
Seção IV (pré/pós em 20 times ágeis, dataset público do Kaggle — dados de terceiros, proveniência não verificável): liderança servidora associada a moral +15,12 pontos e menor tempo de integração de IA — "a 15.12-point average improvement in team morale, a reduction of 11.21 days in AI integration time, and greater emotional well-being and operational efficiency".

\+ **Hasan et al. (2021) — A Survey-based qualitative study to characterize expectations of software developers from five stakeholders**
International Symposium on Empirical Software Engineering and Measurement · termo no artigo: "People skill"
Seção 4.2.2 (expectativas em relação ao gerente): "Participants (48.57%) want their manager to have people skill implying that managers need to understand and guide their teammates. R2,24 expected "To guide us to solve our problems and also think about what might impact our career""; "43.81% of the participants expect managers to have proper accountability, clear vision, reasonable deadline, etc."

+ **Kuutila et al. (2021) — Individual differences limit predicting well-being and productivity using software repositories: a longitudinal industrial study**
Empirical Software Engineering · termo no artigo: “more facilitating and supportive”
Seção 4.4 (entrevistas semiestruturadas com o gerente de projeto e dois desenvolvedores): “The project manager described their leadership style to be more facilitating and supportive. In practice, it meant that nobody was ever assigned to specific tasks, but that the developers chose their tasks from a list for the next sprint”; “Both developers expressed that the project team had plenty of independence for making decisions and that the employer did not intervene in day to day decisions”.

+ **Madampe et al. (2023) — A Framework for Emotion-Oriented Requirements Change Handling in Agile Software Engineering**
IEEE Transactions on Software Engineering · termo no artigo: “prompt and honest communication”
Seção V.C (RQ3.3): “Manager’s prompt and honest communication, better coordination, and having conversations with the team tend to make the practitioners feel both high2 (e.g.: inspired) and high1 (e.g.: satisfied, relaxed) emotions that allow them to better handle the RCs”. βP153: “great project managers are great communicators with high emotional intelligence.”

+ **Russo et al. (2024) — Understanding Developers Well-Being and Productivity: A 2-year Longitudinal Analysis during the COVID-19 Pandemic**
ACM Transactions on Software Engineering and Methodology · termo no artigo: “management support”
Seção 4.5 (análise temática, Wave 6): “Management support is considered a key enabler of hybrid work. When organizations can involve developers and create a sense of belonging, informants are optimistic about the future outlook of hybrid work”. Fechamento da Seção 4.5: “company culture and management support is a clear enabler or inhibitor for a hybrid transition”.

+ **Thasipan et al. (2024) — Determinants of Employee Engagement in the Post-Pandemic Working-from-Home Contexts: Evidence from the Sri Lankan Software Industry**
Proceedings - International Research Conference on Smart Computing and Systems Engineering, SCSE 2024 · termo no artigo: “supervisor's role”
Seção V.C (efeito direto SR→EE 0.425, p<.001; moderação H6 rejeitada): “While supporting the main effects of employee well-being and the supervisor's role on engagement, no significant interaction effect was observed”. Seção V.E: “Investing in supervisor training to support and manage remote teams is crucial, understanding their role indirectly affects engagement through employee well-being”.

+ **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “how their immediate managers supported them”
Seção 4.1.2 (P8): “I have a weekly one on one with my manager. So, just, that’s also how I can try and manage stress”; (P3) o gestor “always give [them] a few days off” quando o participante expressava preocupações de bem-estar. Seção 4.2.1: “Many participants felt that they had a supportive team, but an unsupportive organization”.

+ **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “having a supportive manager”
Seções 4.2.1-4.2.2: gestores apoiadores mitigaram sentimentos de incompetência e criaram segurança psicológica, enquanto microgerenciamento gerou desamparo (P17, P21, P24) — “having a supportive manager can mitigate the feeling of incompetence resulting from work performance concerns”.

– **Graziotin et al. (2017) — On the unhappiness of software developers**
ACM International Conference Proceeding Series · termo no artigo: “Bad decision making”
Tabela 2 e Seção 4 (1.318 respondentes): causa nº 8 (42 menções) — “developers are more often affected (emotionally as well) by bad technical decisions made by their superiors or peers. [...] Bad decision making is also perceived by developers if they are not involved in decision making processes”.

Insegurança no emprego

6 artigos · 1 positiva · 5 negativas

> Medo de perder o emprego: onda de demissão no setor, colega desligado sem explicação, avaliação de performance usada como ameaça, boato de corte. Atinge fortemente quem ficou, e não só quem saiu, e muda o que as pessoas se sentem seguras para dizer.

\+ **Valovy e Buchalcevova (2023) — The Psychological Effects of AI-Assisted Programming on Students and Professionals**

Proceedings - 2023 IEEE International Conference on Software Maintenance and Evolution, ICSME 2023 · termo no artigo: "No fear of replacement"

Tabela I (tema Safety; amostra profissional). P3: "Replacement should also not be a problem because if you learn how to work with AI successfully, you will never be replaced by it." P1: "I was really impressed and happy when I started to use AI because I always dreamt about using something like this. There was no anxiety or relief. The word 'excitement' would describe my feelings best."

– **Hasan et al. (2021) — A Survey-based qualitative study to characterize expectations of software developers from five stakeholders**

International Symposium on Empirical Software Engineering and Measurement · termo no artigo: "job stability"

Seção 1 e 4.5 (expectativas em relação a governos; "Policy and Facility" com 56 códigos em 41 respondentes): "Developers also expect their governments to regulate policy to improve their job stability and from educational institutions to adopt industry-oriented teaching".

– **Obi et al. (2025) — Identifying Factors Contributing to "Bad Days" for Software Developers: A Mixed-Methods Study**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "People and teammates close to you affected by events like layoffs"

Tabela II (rank 6, média 3.96): "People and teammates close to you affected by events like layoffs, covid, etc."

– **Thasipan et al. (2024) — Determinants of Employee Engagement in the Post-Pandemic Working-from-Home Contexts: Evidence from the Sri Lankan Software Industry**

Proceedings - International Research Conference on Smart Computing and Systems Engineering, SCSE 2024 · termo no artigo: "Job safety and security"

Seção V.E (162 respostas válidas, PLS-SEM, H2 aceita com coeficiente 0.569): "The significant positive impact of job safety and security on employee engagement suggests the need for transparent communication about the stability of remote work arrangements. Clear policies and communication regarding job security contribute to a sense of stability, positively influencing engagement".

– **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**

Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "concerns over performance and job security"

Seção 4.1.1 (P14): "No one has said anything – but I worry a lot about like, am I going to have like a negative performance review and be fired".

– **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**

Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "job insecurity"

Seção 4.1.2: demissões fizeram participantes perderem a rede de apoio e temerem pelo emprego (P11, P23) — "participants lost the feeling of relatedness and felt less competent because of the increased workloads and job insecurity".

Inclusão e representatividade

5 artigos · 5 positivas · 0 negativas

> Sentir-se parte do grupo e ver gente parecida com você em posição de decisão. Onde falta, o pertencimento fica permanentemente em dúvida e cada atrito comum é lido como possível sinal de exclusão, o que consome energia que ninguém mais precisa gastar.

+ **Guenes et al. (2025) — Impostor Phenomenon among Software Engineers: Investigating Gender Differences and Well-Being**
Proceedings - 2025 IEEE/ACM 6th Workshop on Gender Equality, Diversity, and Inclusion in Software Engineering, GEICSE 2025 · termo no artigo: “Historical marginalization and systemic exclusion in the computing field”
RQ1 e Discussão: “marginalized gender groups may be disproportionately affected by systemic and cultural factors contributing to IP”.

+ **Trinkenreich et al. (2023) — A Model for Understanding and Reducing Developer Burnout**
Proceedings - International Conference on Software Engineering · termo no artigo: “Inclusiveness”
Seção V.A (3.281 desenvolvedores): “Inclusiveness (H6, B=.12) have a positive and significant association with Work Satisfaction”, $p < 0.001$, e satisfação reduz burnout (H7, B=−0.29). Seção II: “When feeling included by the team, employees believe they are valued for their unique personal characteristics and recognized as important members of the organization”.

+ **Trinkenreich et al. (2024) — Unraveling the Drivers of Sense of Belonging in Software Delivery Teams: Insights from a Large-Scale Survey**
Proceedings - International Conference on Software Engineering · termo no artigo: “lack of representation in leadership positions”
Seção 5.2.3 (análise multigrupo, 9.285 não-líderes): “Women who are not leaders (B =-.121) showed significantly lower feelings of belonging to the team than leaders”. Seção 6.1.3: “the lack of representation in leadership positions could contribute to a more pronounced sense of exclusion or marginalization for women within the organization”.

+ **Trinkenreich et al. (2025) — Guidelines for Cultivating a Sense of Belonging to Reduce Developer Burnout**
IEEE Software · termo no artigo: “inclusiveness”
Tabela 1 (Estudo 1, 3.281 respostas; ICSE-SEIP 2023): “Organizational culture, climate for learning, sense of belonging, and inclusiveness are positively associated with work satisfaction, which in turn is associated with reduced burnout”. O achado de gênero do Estudo 3 está contabilizado no artigo Trinkenreich20242345.

+ **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “feel included among other software developers”
Seção 4.2.3 (P13, P25): “whether participants could identify (e.g., gender and race) with others at work and feel included among other software developers”.

Cultura autoritária

4 artigos · 0 positivas · 4 negativas

> Quanto a hierarquia pesa no dia a dia: decisão que só desce pronta, distância para chegar em quem decide, discordância lida como insubordinação. O oposto aparece como estrutura mais plana e acesso direto a quem tem contexto, e muda quem se sente autorizado a apontar problema.

– **Khalid et al. (2022) — Relationship between Organization's Physical and Psychological Environment and Employees' Mental Satisfaction: An Empirical Analysis of Employee Turnover in Software Industry**

2022 16th International Conference on Open Source Systems and Technologies, ICOSST 2022 - Proceedings · termo no artigo: "strict top-down decision making pattern"

Seção V.B (97 profissionais de CS/TI de software houses de Lahore): "Although, there were a few organizations which were following a consensus-based strategy, but the majority were following a strict top-down decision making pattern"; "The CEOs, on the other hand, were supportive towards good ideas too but insistent on being an authoritative personnel in their organization(s) when it comes to interacting with their employees. This difference can affect the level and ease of communication between managerial and non-managerial employees".

– **Trinkenreich et al. (2024) — Unraveling the Drivers of Sense of Belonging in Software Delivery Teams: Insights from a Large-Scale Survey**

Proceedings - International Conference on Software Engineering · termo no artigo: "Power Distance"

Seção 6.1.3 (variável de controle, B=-.071, p<.05): "team belonging is positively associated with organizational tenure and negatively associated with being a woman (gender minority) and authoritativeness (power distance). [...] In contrast, higher levels of power distance in national culture are associated with lower feelings of belonging to the team".

– **Trinkenreich et al. (2025) — Guidelines for Cultivating a Sense of Belonging to Reduce Developer Burnout**

IEEE Software · termo no artigo: "Authoritative Culture"

Tabela 1 (Estudo 2, 318 contribuidores do kernel Linux): "Power distance negatively affects the sense of virtual community". O achado do Estudo 3 (10.781 respostas) está contabilizado no artigo Trinkenreich20242345.

– **Weerasekara et al. (2020) — ICT supported occupational stress management: The views of hr managers in Sri Lankan software companies**

Proceedings of the 12th IADIS International Conference e-Health 2020, EH 2020 - Part of the 14th Multi Conference on Computer Science and Information Systems, MCCSIS 2020 · termo no artigo: "open-door policy"

Seção 5.1.3: "The companies claimed to follow a flat structure and to have a friendly atmosphere inside the organization. They followed the open-door policy, to have the employees feel easy and welcomed to meet any other employee or manager without any prior appointment". HR3: "We have no hierarchy; we have a flat structure here."

Monitoramento no trabalho

4 artigos · 2 positivas · 2 negativas

> Medição e vigilância do trabalho: métrica individual de commit ou de ticket, tempo online, relatório de atividade, painel comparando pessoas. Muda o comportamento antes de medir qualquer coisa — passa-se a otimizar o indicador, e a pressão por reportar consome tempo do trabalho reportado.

+ **Hicks et al. (2024) — Developer Thriving: Four Sociocognitive Factors That Create Resilient Productivity on Software Teams**

IEEE Software · termo no artigo: "healthy metrics use"

Resultados (Tabela 8: HMU→DTS β=0.12, p<0.01; HMU→VVQ β=0.32, p<0.001): "HMU was associated with greater visibility and value of work, highlighting HMU as one factor for creating visibility and value of software work"; a segunda pergunta do construto exigia que o time "measure the right things".

+ **Leme et al. (2024) — Mental Health and Productivity in Software Development: A Study with the Bravo Central Platform**
ACM International Conference Proceeding Series · termo no artigo: “monitoramento de saúde mental”
Seção 5 (6 membros do Studio Bravarda, 16 semanas): “Essas análises foram capazes de gerar conversas valiosas sobre pontos fracos na saúde mental da equipe. Isso possibilitou a observação isolada de certos membros que estavam com algum ponto da sua saúde mental abaixo da média da equipe, ou abaixo do que é considerado saudável pelo questionário”. Seção 5.1: “A reflexão coletiva resultou em estratégias para melhorar a distribuição de tarefas, garantindo que todos os membros tenham um volume de trabalho adequado às suas capacidades e disponibilidade”.

– **Arriel et al. (2026) — Understanding developer well-being: measuring mental health and productivity in software teams**
Empirical Software Engineering · termo no artigo: “monitoring and performance measurement in the workplace”
Seções 6.2.1-6.2.2 (P6, P7, P13, P20): “a natural response to the perception of privacy invasion, reduced autonomy, and increased stress that often accompany the implementation of electronic monitoring systems”.

– **Elsalmy et al. (2025) — Exploring Gamification Risks and Implications in Agile Software Development Enterprises: An Empirical Study**
ICSIE 2024 - 2024 13th International Conference on Software and Information Engineering · termo no artigo: “effort misinterpretation”
Seções 4.1 e 4.4: pontuação baseada em commits e linhas de código levou a esforço mal interpretado, julgamento só pelos números e trapaça no sistema (P2, P12) — “I earn points on x no of commits per day [...] and get judged on them only and management never involves us”.

Reconhecimento no trabalho

4 artigos · 4 positivas · 0 negativas

> Trabalho ser visto e valorizado: crédito na entrega, feedback que existe, avaliação que acontece, e visibilidade para o que é estruturalmente invisível — refatorar, revisar código, apagar incêndio, sustentar sistema legado. Trabalho invisível bem feito não gera reconhecimento nenhum, e isso é percebido.

+ **Hicks et al. (2024) — Developer Thriving: Four Sociocognitive Factors That Create Resilient Productivity on Software Teams**
IEEE Software · termo no artigo: “visibility and value of work”
Resultados (survey, 1.282 devs; mediação serial, Tabela 8: VVQ→DTS β=0.65, p<0.001): “healthy team metrics use and greater visibility and value of software work were both significantly associated with greater developer thriving, with visibility and value of work having the stronger effect on thriving”.

+ **Trinkenreich et al. (2024) — Unraveling the Drivers of Sense of Belonging in Software Delivery Teams: Insights from a Large-Scale Survey**
Proceedings - International Conference on Software Engineering · termo no artigo: “Work Appreciation”
Seção 6.1.1 (10.781 respondentes, PLS-SEM, H1 aceita): “Work Appreciation was positively associated with belonging to the team (H1, B=.392). Achieving a satisfactory equilibrium between contributions and rewards (WA1), being recognized by leadership (WA2), and perceiving own contribution with a meaningful impact on the company (WA3) are positively associated with belongingness to the team”.

+ **Trinkenreich et al. (2025) — Guidelines for Cultivating a Sense of Belonging to Reduce Developer Burnout**
IEEE Software · termo no artigo: “work appreciation”
Tabela 1 (Estudo 3, 10.781 respostas) e seção Timely and Consistent Recognition: “Positive leadership feedback is crucial for acknowledging employees, fostering belongingness, and reducing burnout”.

+ **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: “not recognized for the work”
Seção 4.1.1 (P8): “P8 expressed that when they were “not recognized for the work that [they] did. Like, that piss [them] off””.

Iniciativas superficiais de bem-estar

3 artigos · 0 positivas · 3 negativas

> Ação de bem-estar que não mexe na causa: palestra de mindfulness, dia da saúde mental, kit de autocuidado, enquanto prazo, carga e cobrança seguem iguais. Costuma produzir cinismo e desconfiança da intenção da empresa — nesses casos, o efeito é pior que não ter feito nada.

– **Arriel et al. (2026) — Understanding developer well-being: measuring mental health and productivity in software teams**

Empirical Software Engineering · termo no artigo: "superficial or performative interventions"

Seção 6.1.3: programas de saúde mental percebidos como simbólicos ou de fachada geraram ceticismo em vez de bem-estar (P3, P8) — "corporate mental health programs often exist as symbolic measures rather than meaningful strategies to improve employee well-being".

– **Cardenas-Castro et al. (2019) — Soft skills training: Performance psychology applied to software development**

Proceedings - 2019 IEEE/ACM 12th International Workshop on Cooperative and Human Aspects of Software Engineering, CHASE 2019 · termo no artigo: "pre-designed talks or workshops"

Seção IV: "Another important finding from this exercise is that the context defines the intervention process and considering it can produce higher value and impact in comparison with pre-designed talks or workshops. [...] the training sessions and programs must be flexible enough in order to facilitate its adaptation to the context: type of client and project, project times, team maturity, among others". Ainda na Seção IV: "each participant has its own initial needs and skills and embraces the information given at its own pace, which is a factor commonly overlooked in a group training".

– **Wong et al. (2023) — Mental Wellbeing at Work: Perspectives of Software Engineers**

Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "skeptical towards the organizations' intentions of implementing mental wellbeing policies"

Seção 4.2.1 (P1): "[organizations] can kind of disguise" employer-focused policies (e.g., data-tracking and money-saving) as mental wellbeing support for employees"; (P4) sobre PTO: "while you could work on getting time off...the work was still there when you came back. And so when you came back, you felt more stressed because then you were catching up"; (P2): "will state my paranoia that the company is paying for that [hotline] and therefore, how do we know where the reporting is going to".

Senso de propósito

3 artigos · 3 positivas · 0 negativas

> Ver sentido no que se faz e alguma conexão entre o próprio trabalho e o que a empresa busca. Sem isso, o trabalho vira execução de tarefa, e a ligação com a organização — o sentir-se parte de algo — desaparece, o que aparece cedo em intenção de sair.

+ **Brandebusemeyer et al. (2025) — Wearables to measure developer experience at work**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: "lack of team/organization vision"

Seção IV.F e Tabela III: a falta de visão de time/organização é o 2º maior desafio, e o maior desejo de melhoria (45% dos 20) é comunicação aberta de metas e visão de projeto — "clear and open discussion and communication of project goals and project vision".

+ **Khalid et al. (2022) — Relationship between Organization's Physical and Psychological Environment and Employees' Mental Satisfaction: An Empirical Analysis of Employee Turnover in Software Industry**
2022 16th International Conference on Open Source Systems and Technologies, ICOSST 2022 - Proceedings · termo no artigo: "missing the strong connection one feels in being a part of an organization"
Seção V.B (empregados): "the employees were very confident about their work affecting the company and they felt that their jobs were somewhat challenging to keep them engaged in their work. Yet, the majority were missing the strong connection one feels in being a part of an organization. If they were doing important and enjoyable tasks, the missing connection was then because of something other than the work they were carrying out".

+ **Wong et al. (2025) — 'It's a spectrum': Exploring Autonomy, Competence, and Relatedness in Software Development Processes and Tools**
Conference on Human Factors in Computing Systems - Proceedings · termo no artigo: "alignment with company values and goals"
Seção 4.2.2 (P9, P10, P18, P24): "alignment with company values and goals significantly affected relatedness".

Justiça organizacional

2 artigos · 0 positivas · 2 negativas

> Percepção de que regra, promoção, benefício e cobrança valem igual para todos, e que o critério é explícito. Injustiça percebida desengaja inclusive quem não foi prejudicado, porque o que se aprende é que o resultado não depende do trabalho.

– **Hasan et al. (2021) — A Survey-based qualitative study to characterize expectations of software developers from five stakeholders**
International Symposium on Empirical Software Engineering and Measurement · termo no artigo: "Work Benefits"
Seção 4.1.2 (expectativas em relação às organizações): "Participants (29.27%) expect timely salary, proper salary, and food supply. R2,17 expects "giving salary according to contribution"".

– **Khalid et al. (2022) — Relationship between Organization's Physical and Psychological Environment and Employees' Mental Satisfaction: An Empirical Analysis of Employee Turnover in Software Industry**
2022 16th International Conference on Open Source Systems and Technologies, ICOSST 2022 - Proceedings · termo no artigo: "biased rules"
Abstract: "their loyalty and mental health is being compromised by a number of psychological factors, including biased rules and limited growth opportunities and how this impacts their desire to quit". Seção V.C: "While all the CEOs stated all policies to be fair and justified, some of the employees had other thoughts about them. [...] the minor deviations of biasness were present. These deviations can also be the reason certain employees were feeling disintegrated in their organizations". Sobre remuneração: "The 32% of respondents who opted for better pay indicates that the work to salary ratio in organizations was not justified".

Acomodações para neurodivergência

1 artigo · 1 positiva · 0 negativas

> Ajustes concretos no posto de trabalho para pessoas neurodivergentes — autismo, TDAH, dislexia: fone com cancelamento de ruído, pauta enviada antes da reunião, permissão para responder por escrito em vez de falar ao vivo, horário deslocado. Custam pouco e quase nunca são negados quando pedidos; o problema é que raramente estão oferecidos por padrão, então dependem de a pessoa saber que existem e se expor para pedir.

\+ **Newman et al. (2025) — Disclosure of Neurodivergence in Software Workplaces: a Mixed Methods Study of Forum and Survey Perspectives**

ASSETS 2025 - Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility · termo no artigo: “accommodations”

Seções 5.2 e 5.4 (fórum r/ADHD_Programmers e survey): “Commenters received workplace support through accommodations and shared work style understanding, leading to better task allocation”; “Positive outcomes included improved job satisfaction from accommodations and reduced stress from not having to pass”.

Clima de aprendizagem

1 artigo · 1 positiva · 0 negativas

> Se a empresa dá tempo e espaço reais para aprender dentro do horário de trabalho, ou se estudar é assunto para a noite e o fim de semana. Inclui haver com quem aprender e poder passar tempo numa coisa nova sem que isso conte como hora perdida.

\+ **Trinkenreich et al. (2023) — A Model for Understanding and Reducing Developer Burnout**

Proceedings - International Conference on Software Engineering · termo no artigo: “Climate for Learning”

Seção V.A (3.281 desenvolvedores): “Climate for Learning (H5, B=.22) [...] have a positive and significant association with Work Satisfaction”, que por sua vez tem “a reverse (negative) and significant association with Burnout (H7, B=−0.29)”. Definição na Seção II: “An organization that exhibits a climate for learning makes resources available for continued education and offers continuous encouragement to teams to learn by providing them space and time to acquire new knowledge and explore ideas”.

Conformidade e sigilo de dados

1 artigo · 0 positivas · 1 negativa

> Regras de sigilo e proteção de dados que limitam como se pode trabalhar: não poder colar código em ferramenta de IA, restrição sobre onde o dado transita, receio de responder pessoalmente por um vazamento. O peso não vem só da regra, mas da falta de clareza sobre o que é permitido.

\− **Valovy e Buchalcevova (2023) — The Psychological Effects of AI-Assisted Programming on Students and Professionals**

Proceedings - 2023 IEEE International Conference on Software Maintenance and Evolution, ICSME 2023 · termo no artigo: “Data privacy concerns”

Tabela I (tema Ethical Considerations; amostra profissional). P2: "Where I am a contractor, we have signed some NDA and we cannot share code. So it is forbidden to copy and paste our code to the AI." P5: "Yes, the biggest disadvantage is that it takes the data and learns from it. It poses potential lawsuit threats." P3: "Potential risk: A leak of API keys. It happened in my company."

Cultura organizacional generativa

1 artigo · 1 positiva · 0 negativas

> Cultura que trata incidente e erro como informação, não como caça ao culpado: post-mortem sem procurar responsável, informação circulando entre áreas, más notícias subindo rápido. O efeito prático é problema aparecendo cedo, enquanto ainda é barato.

\+ **Trinkenreich et al. (2023) — A Model for Understanding and Reducing Developer Burnout**

Proceedings - International Conference on Software Engineering · termo no artigo: "Generative Organizational Culture"

Seção V.A (survey com 3.281 desenvolvedores de times de entrega de software da Globant, SEM): "Results revealed that a Generative Organizational Culture has a positive significant association with Sense of Belonging (H1, B=.48), Climate for Learning (H2, B=.54), and Inclusiveness (H3, B=.45)", todas com $p < 0.001$. Definição na Seção II: "Generative organizations are performance-oriented, with good information flow, high levels of cooperation and trust, and bridging between teams. The generative level can be achieved by creating cross-functional teams to improve cooperation, holding blameless postmortems, sharing risks and responsibilities, breaking down silos".

Estigma da neurodivergência

1 artigo · 0 positivas · 1 negativa

> O julgamento que a pessoa neurodivergente percebe no ambiente: ser lida como difícil, desinteressada, sem jeito com gente, ou ter traço confundido com falta de compromisso. O efeito prático é esconder a condição e não pedir o ajuste que resolveria o problema.

– **Newman et al. (2025) — Disclosure of Neurodivergence in Software Workplaces: a Mixed Methods Study of Forum and Survey Perspectives**

ASSETS 2025 - Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility · termo no artigo: "perceived stigma"

Seção 5.4 (survey, participantes ND): "For ND participants, perceived stigma was infrequent (the median frequency was "Never"), but negatively related with PMH"; "Perceived workplace stigma decreased mental health".

Estigma do uso de substâncias

1 artigo · 0 positivas · 1 negativa

> Julgamento sobre quem usa substância psicoativa, incluindo medicação prescrita para TDAH, ansiedade ou depressão. Impede falar do assunto com gestor e com saúde ocupacional, e portanto impede pedir apoio — o silêncio é consequência do estigma, não do uso.

– **Newman et al. (2023) — From Organizations to Individuals: Psychoactive Substance Use By Professional Programmers**

Proceedings - International Conference on Software Engineering · termo no artigo: "stigma around psychoactive substance use"

Seções VIII-B e IX-C (13 de 26 participantes): "open disclosure of some substances is not common in the corporate cultures of our participants"; "The most requested change (13 participants) would be decreasing stigma toward psychoactive substance use in software"; "I think it should be normalized if someone takes stress or antidepressants" (P6).

Oportunidades de carreira

1 artigo · 1 positiva · 0 negativas

> Ter caminho visível de crescimento: trilha com critério claro, promoção que efetivamente acontece, acesso a projeto que agrega. Onde não existe, sair da empresa passa a ser o único mecanismo de progressão disponível, e as pessoas agem de acordo.

\+ **Hasan et al. (2021) — A Survey-based qualitative study to characterize expectations of software developers from five stakeholders**

International Symposium on Empirical Software Engineering and Measurement · termo no artigo: "Career Opportunities"

Seção 4.1.2 e 4.2.2 (54 códigos em 47 respondentes ao todo): "21.95% of the participants expect various career opportunities including career growth and security, proper structure from the organization"; R2,2: "Introduce a proper structure which will benefit both employees and the organization". Conclusão: "Out of the 18 categories, three categories are related to career growth. This observation supports previous research that happiness cannot be assured by simply offering more money or a promotion".

Política antidrogas

1 artigo · 0 positivas · 1 negativa

> Regra e teste sobre uso de substância no trabalho, incluindo teste aleatório. Afeta também quem usa medicação prescrita, que precisa se justificar, e o próprio receio de ser testado tem peso independente de haver qualquer uso.

– **Newman et al. (2023) — From Organizations to Individuals: Psychoactive Substance Use By Professional Programmers**

Proceedings - International Conference on Software Engineering · termo no artigo: "anti-drug policies"

Seções IX-A e IX-B (26 programadores profissionais): "potential tests do lead to additional stress and frustration"; "more restrictive drug policies, especially those that admit the possibility of random testing, are more likely to cause substance-using programmers to not apply to work at a company"; "would not be comfortable working somewhere where they were going to . . . have that little trust in me" (P20).

Preparação organizacional para crises

1 artigo · 1 positiva · 0 negativas

> Quão pronta a empresa está para evento fora do normal: pandemia, incidente grave, mudança abrupta de modo de operação, perda súbita de pessoas-chave. Quando não há preparo, a adaptação é improvisada e o custo dela é cobrado da equipe, em horas e em incerteza.

\+ **Ralph et al. (2020) — Pandemic programming: How COVID-19 affects software developers and how their organizations can help**

Empirical Software Engineering · termo no artigo: "Disaster preparedness"

Seção 5.5 (SEM; H5 e H10 suportadas): "change in perceived productivity depends on home office ergonomics and disaster preparedness [...] and disaster preparedness is inversely related to fear". Seção 4: "Disaster preparedness is the degree to which a person is ready for a natural disaster".

Silêncio organizacional

1 artigo · 0 positivas · 1 negativa

> Se as pessoas falam ou se calam diante de problema e discordância. O silêncio aqui não é timidez: é comportamento aprendido, resultado de já se ter falado antes e nada ter acontecido, ou de ter havido custo. O efeito é a organização perder justamente a informação que só quem executa tem.

– **Sánchez-Gordón et al. (2026) — The role of voice and silence behaviors in software development: a structural equation modeling analysis**

Information and Software Technology · termo no artigo: "silence behaviors"

Tabela de resultados do SEM (158 profissionais de software), H5 aceita com estimativa 0.356: "Change in the software developer's silence is directly and significantly related to withdrawal (H5) showing a stronger association than his/her voice (H7), which is negatively related but not significant". O desfecho withdrawal é a dimensão de burnout medida no modelo. H6 aceita para o outro polo: "Work performance is directly and significantly related to the software developer's voice (H6) whereas silence (H8) is negatively related but not significant".

Treinamento de soft skills

1 artigo · 1 positiva · 0 negativas

> Formação em comunicação, mediação de conflito, feedback e regulação emocional. É raro na área, que costuma treinar só o técnico, e a falta aparece de forma concreta em conflito que escala sem necessidade e em feedback que machuca sem corrigir nada.

+ **Cardenas-Castro et al. (2019) — Soft skills training: Performance psychology applied to software development**

Proceedings - 2019 IEEE/ACM 12th International Workshop on Cooperative and Human Aspects of Software Engineering, CHASE 2019 · termo no artigo: "Soft skills training"

Seção IV (140 funcionários em 6 times de uma empresa de software na Colômbia, 69 no diagnóstico): "The qualitative reports of the participants show an overall improvement in the skills worked and in the general feeling of well-being. Nonetheless, this report requires additional measurements that allow identifying more clearly the impact of the treatment". Seção III.C.1: as necessidades diagnosticadas foram "coping strategies (in particular high levels of self-criticism combined with wishful thinking), communication (assertiveness, fear to speak in public), time management (procrastination and the limits of effort), emotional regulation".

Individual

9 fatores · 13 artigos distintos

Alinhamento do uso do tempo

5 artigos · 4 positivas · 1 negativa

> Diferença entre como a pessoa gostaria de dividir a semana e como ela de fato vai. O caso típico é quem foi contratado para programar e passa o dia em reunião, suporte e desbloqueio de outros. Quanto maior a distância entre a semana ideal e a real, pior o resultado — mesmo quando a carga total não é alta.

\+ **Kumar et al. (2025) — Time Warp: The Gap Between Developers' Ideal vs Actual Workweeks in an AI-Driven Era**

IEEE/ACM International Conference on Software Engineering - Software Engineering in Practice · termo no artigo: “the gap between these two workweeks widens”

Abstract e Seção IV (survey, 484 devs da Microsoft): “as the gap between these two workweeks widens, we observe a decline in both productivity and satisfaction”.

\+ **Meyer et al. (2021) — Enabling good work habits in software developers through reflective goal-setting**

IEEE Transactions on Software Engineering · termo no artigo: “Balance Coding Time”

Seção 4.2 (10 desenvolvedores): “Another reason why developers want to avoid deviations form work is to ensure they can spend enough time with coding, by balancing activities and setting limits (S11) or by blocking a specific amount of time for coding in the calendar”; “The importance of finding a good balance between coding and other work was recently shown to increase developers’ job satisfaction [50]”.

\+ **Russo et al. (2021) — Predictors of well-being and productivity among software professionals during the COVID-19 pandemic – a longitudinal study**

Empirical Software Engineering · termo no artigo: “daily routines”

Seção 5 (estudo longitudinal de duas ondas, N=192; Tabela 2, preditores de bem-estar na onda 1): “At time 1, stress (negatively), social contacts, and daily routines uniquely predicted well-being at α = .05”. Tabela 2: Daily routines B = 0.125, p = 0.043.

\+ **Russo et al. (2021) — Predictors of well-being and productivity among software professionals during the COVID-19 pandemic --- a longitudinal study**

Empirical Software Engineering · termo no artigo: “daily routines uniquely predicted well-being”

Seção 5.2 (N = 192, onda 1): “At time 1, stress (negatively), social contacts, and daily routines uniquely predicted well-being at α = .05”.

– **Kuutila et al. (2021) — Individual differences limit predicting well-being and productivity using software repositories: a longitudinal industrial study**

Empirical Software Engineering · termo no artigo: “the feeling of hurry”

Seção 4.4 (Developer 1): “Well, I believe, in this project, the feeling of hurry, has been precisely that you don’t have time to develop, but 70% of your time is going to everything else. When you are not in a hurry, you have seven and a half hours to code”. Consequência relatada: “When I feel like I have to get something done... I don’t partake in internal educational events or other training”.

Práticas de mindfulness

4 artigos · 4 positivas · 0 negativas

> Prática deliberada e mantida de atenção e respiração, individual ou incorporada à rotina do time. Não se confunde com ação performática de bem-estar: aqui o que aparece na evidência é prática continuada, com efeito medido — o que também significa que exige tempo protegido para acontecer.

+ **Bernardez et al. (2023) — On the Impact and Lessons Learned from Mindfulness Practice in a Real-World Software Company**
International Symposium on Empirical Software Engineering and Measurement · termo no artigo: "continued mindfulness practice"
Seção V.B (experimento controlado na INPRO, 19 tratamento + 19 controle, 6 semanas de prática diária): "employees in the Exp group achieved significantly higher levels of attention awareness after the intervention compared to the Ctrl group, as described by the moment *group interaction (p = 0.031)"; "the test (Table VI) revealed a significant decrease (p=0.016) in the levels of techno–stress expericenced by employees in the Exp group compared to those in the Ctrl group, with a large effect size (η2p = 0.150)". Seção V.D: "regarding the perceived benefits observed after six weeks of continued mindfulness practice (I–K) in terms of concentration, being in the present and improving sleep quality, almost half of the responses fall into the Agree category".

+ **Jafarzadeh et al. (2025) — Burnout in agile teams: The role of mindful software development**
Information and Software Technology · termo no artigo: "mindful engagement with agile methods"
Seção 3 e 5.4 (319 profissionais ágeis): H1 "Mindful engagement with agile methods is positively associated" com coping focado no problema, sustentada a α = 0.001 — e via esse caminho reduz burnout. Definição na Seção 2: "we define agile mindfulness as a dynamic agile-specific trait, evident when working with agile methodologies, whereby the agile team members focus on the present, pay attention to detail, exhibit a willingness to consider other uses, and express genuine interest in investigating features and failures of agile methods".

+ **Montes et al. (2024) — Qualifying and Quantifying the Benefits of Mindfulness Practices for IT Workers**
Proceedings - 2024 10th International Conference on ICT for Sustainability, ICT4S 2024 · termo no artigo: "Mindfulness Practices"
Discussão e Conclusão (intervenção de 8 semanas; 610 entradas de diário, 65 participantes): "The average daily rating for all participants was increased from 6.86 (week 1) to 7.61 (week 8). This meant that the overall increase was 10.95%"; "breathing practices in combination with reflective journaling have a positive impact on IT workers' well-being and productivity, but further research is needed to establish it as a fact".

+ **Penzenstadler et al. (2022) — Take a deep breath: Benefits of neuroplasticity practices for software developers and computer workers in a family of experiments**
Empirical Software Engineering · termo no artigo: "breathing practice"
Abstract e Seção 5 (família de experimentos Rise2Flow 1 e 2; 34 e 33 participantes concluintes, 1656 registros de diário): "The intervention showed improvements in participants' experienced inner states despite an ongoing pandemic and intense outer circumstances for most. Over the course of the study, we found an improvement in the participants' ratings of how often they found themselves in good spirits as well as in a calm and relaxed state"; "The data indicates usefulness and effectiveness of an intervention for computer workers in terms of increasing well-being and resilience".

Qualidade do sono

3 artigos · 3 positivas · 0 negativas

> Quanto e como se dorme, e o quanto o trabalho estraga isso: deploy de madrugada, plantão, cabeça que não desliga depois de um incidente, fuso de time distribuído. Sono ruim volta para o trabalho como erro, irritação e decisão pior, fechando o ciclo.

\+ **Leme et al. (2024) — Mental Health and Productivity in Software Development: A Study with the Bravo Central Platform**

ACM International Conference Proceeding Series · termo no artigo: "qualidade do sono"

Seção 5.2 (6 membros do Studio Bravarda): "A frequência com que os desenvolvedores acordam se sentindo restaurados e descansados está diretamente relacionada à qualidade do sono e, por conseguinte, à sua produtividade naquele dia. Portanto, a análise resultou em uma conversa sobre ciclo circadiano"; "Eventos pessoais: mudanças na vida cotidiana, responsabilidades familiares ou eventos externos impactaram a qualidade do sono e a sensação de energia durante o dia".

\+ **Russo et al. (2021) — Predictors of well-being and productivity among software professionals during the COVID-19 pandemic – a longitudinal study**

Empirical Software Engineering · termo no artigo: "quality of sleep"

Seção 5 (Tabela 4, preditores de bem-estar na onda 2): "At time 2, need for competence and autonomy, stress, quality of social contacts, and quality of sleep uniquely predicted well-being at α = .05". Tabela 4: Quality of Sleep B = 0.144, p = 0.011.

\+ **Russo et al. (2021) — Predictors of well-being and productivity among software professionals during the COVID-19 pandemic --- a longitudinal study**

Empirical Software Engineering · termo no artigo: "quality of sleep uniquely predicted well-being"

Seção 5.2 (N = 192, onda 2): "need for competence and autonomy, stress, quality of social contacts, and quality of sleep uniquely predicted well-being at α = .05".

Contatos sociais fora do trabalho

1 artigo · 1 positiva · 0 negativas

> *Sem descrição no catálogo (pendência da tarefa 1.3).*

\+ **Russo et al. (2021) — Predictors of well-being and productivity among software professionals during the COVID-19 pandemic --- a longitudinal study**

Empirical Software Engineering · termo no artigo: "the quality of social contacts predicted positively"

Seção 5.2 (N = 192, ambas as ondas): "Together, stress and quality of social contacts predicted at both time points significantly well-being".

Coping focado na emoção

1 artigo · 0 positivas · 1 negativa

> Como a pessoa lida com o estresse mexendo no que sente, e não na causa: desabafar com um colega, se distrair, se afastar da ferramenta por um tempo, controlar a reação. Alivia no curto prazo e às vezes é a única opção disponível, mas o problema que gerou o estresse continua lá.

– **Jafarzadeh et al. (2025) — Burnout in agile teams: The role of mindful software development**
Information and Software Technology · termo no artigo: "Emotion-focused coping strategies"
Seção 5.4 e Discussão (319 profissionais ágeis): H4 "Emotion-focused coping strategies increase work exhaustion from using agile methods" foi sustentada — "EFC efforts, which in turn are positively associated with burnout ($\beta = 0.53$, $p < 0.001$)". Achado contraintuitivo em H2: "individuals' mindful engagement with agile methods may also be linked to increased EFC efforts", com "they try to gain emotional stability by venting or seeking emotional support".

Coping focado no problema

1 artigo · 1 positiva · 0 negativas

> Lidar com o estresse atacando a causa: consertar a ferramenta que atrapalha, mudar a rotina de uso, renegociar prazo ou escopo, achar um contorno que funcione. Só está disponível para quem tem alguma margem real de ação — sem autonomia, a pessoa é empurrada para as estratégias emocionais.

+ **Jafarzadeh et al. (2025) — Burnout in agile teams: The role of mindful software development**
Information and Software Technology · termo no artigo: "Problem-focused coping strategies"
Seção 5.4 (survey com 319 profissionais de TI e desenvolvimento de software que usam métodos ágeis, path analysis em SmartPLS): H3 "Problem-focused coping strategies reduce work exhaustion associated with use of agile methods" — "The relationships between the constructs were all statistically significant (at $\alpha = 0.001$ for H1, H3 and H4)".

Divulgação da neurodivergência

1 artigo · 1 positiva · 0 negativas

> A decisão de contar ou não no trabalho que é neurodivergente. Contar pode destravar acomodação e explicar comportamento mal interpretado; também expõe a estigma e a julgamento sobre competência. O peso está na decisão em si, que se repete a cada time novo e a cada gestor novo.

+ **Newman et al. (2025) — Disclosure of Neurodivergence in Software Workplaces: a Mixed Methods Study of Forum and Survey Perspectives**
ASSETS 2025 - Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility · termo no artigo: "disclosure of neurodivergence"
Seções 5.3 e 5.4 (survey com 493 engenheiros de software, 299 neurodivergentes): "219 ND participants (85%) had positive or neutral disclosure outcomes"; "We found a relationship between positive disclosure outcomes from disclosing to a manager and job satisfaction ($\chi 2$, $p = .004$)"; "I'm much happier in a workplace that recognizes and works with my ADHD rather than me having to mask non-stop".

Fenômeno do impostor

1 artigo · 0 positivas · 1 negativa

> Sensação persistente de não ser competente de verdade e de que isso vai ser descoberto, mesmo com evidência clara em contrário. Aparece com força em quem entra num time mais sênior, muda de stack ou é o único com determinado perfil no grupo.

– **Guenes et al. (2025) — Impostor Phenomenon among Software Engineers: Investigating Gender Differences and Well-Being**

Proceedings - 2025 IEEE/ACM 6th Workshop on Gender Equality, Diversity, and Inclusion in Software Engineering, GEICSE 2025 · termo no artigo: "Impostor Phenomenon"

RQ2 (survey com 624 profissionais, CIPS + WHO-5): 52,7% sofrem IP frequente/intenso, e quem sofre tem bem-estar significativamente menor (Mann-Whitney p<2.49e-11 homens, p<4.61e-05 mulheres), com medianas abaixo do limiar de bem-estar pobre e o dobro de casos na faixa de depressão maior — "software professionals with IP consistently show significantly lower subjective well-being than their counterparts without IP".

Uso de substâncias psicoativas

1 artigo · 1 positiva · 0 negativas

> Uso de substância que altera cognição ou humor — de café e estimulante prescrito a álcool e outras drogas — na relação com demanda de foco, pressão de prazo e sono ruim. Interessa aqui como resposta a condições de trabalho, não como característica pessoal.

+ **Newman et al. (2023) — From Organizations to Individuals: Psychoactive Substance Use By Professional Programmers**

Proceedings - International Conference on Software Engineering · termo no artigo: "psychoactive substance use"

Seções VI-A e VI-B (26 programadores profissionais, P3 e P5): "mental health is a primary driver of psychoactive substance use when developing software"; "Of course, it affects my software work. If I do have depression, I cannot work" (P3); "the impact of prescription stimulants on their software work is almost uniformly positive: 'Oh, it's been almost life-changing. [...] I'm way more focused obviously, but I'm also way more productive'" (P5).

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